

1. Laws of Motion



- Motion
- Distance and displacement
- Acceleration
- Newton's laws of motion and related equations.

Motion of an object



Can you tell?

In which of the following examples can you sense motion?
How will you explain presence and absence of motion?

1. The flight of a bird
2. A stationary train
3. Leaves flying through air
4. A stone lying on a hill

We see the motion of several objects every day. Sometimes we cannot see the motion of an object directly, as in the case of a breeze. Can you list other examples of motion, besides those given here?



Think about it.

1. You are travelling in a bus. Is the person sitting next to you in motion?
2. What do you take into consideration to decide if an object is moving or not?

You have learnt that motion is a relative concept. If the position of an object is changing with respect to its surroundings, then it is said to be in motion. Otherwise, it is said to be at rest.

Displacement and distance

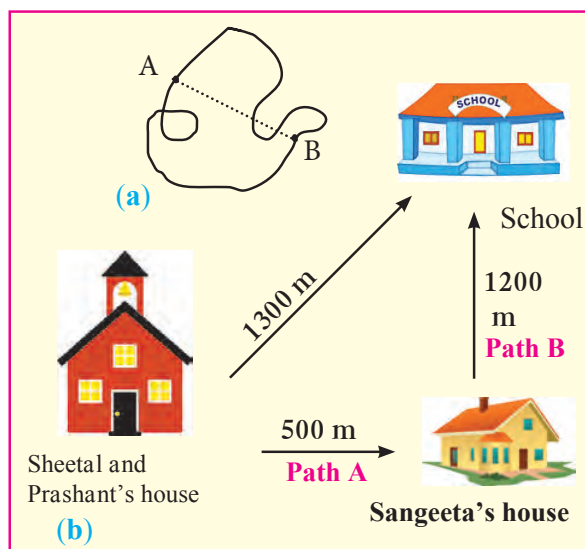


Let's try this

1. Measure the distance between points A and B in different ways as shown in figure 1.1(a)
2. Now measure the distance along the dotted line. Which distance is correct according to you and why?



Think about it.



1.1 Location of the school and houses

- A. Sheetal first went to her friend Sangeeta's house on her way to school (see figure 1.1b).
- B. Prashant went straight from home to school. Both are walking with the same speed. Who will take less time to reach the school and why?

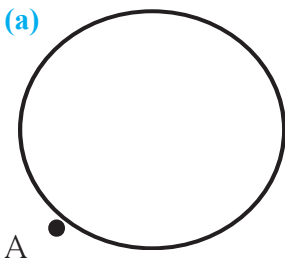
In the above example, is there a difference between the actual distance and the distance travelled? What is it?

‘Distance’ is the length of the actual path travelled by an object in motion while going from one point to another, whereas displacement is the minimum distance between the starting and finishing points.



Use your brain power!

(a)

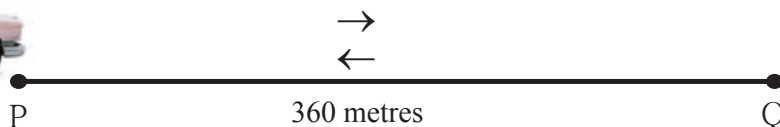


(b)



1. Every morning, Swaralee walks round the edge of a circular field having a radius of 100 m. As shown in figure 1.2 (a), if she starts from the point A and takes one round, how much distance has she walked and what is her displacement?
2. If a car, starting from point P, goes to point Q (see figure 1.2 b) and then returns to point P, how much distance has it travelled and what is its displacement?

1.2 Distance and displacement



Even if the displacement of an object is zero, the actual distance traversed by it may not be zero.

Speed and velocity



Can you recall?

1. What are vectors and scalars?
2. Which of the quantities distance, speed, velocity, time and displacement are scalars and which are vectors?

$$\text{Speed} = \frac{\text{Total distance travelled}}{\text{Time required}}$$

The distance travelled in one direction by an object in unit time is called its velocity. Here, unit time can be one second, one minute, one hour, etc. If large units are used, one year can also be used as a unit of time.

The displacement that occurs in unit time is called velocity.

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$



Always remember

1. The units of speed and velocity are the same. In the SI system, the unit is m/s while in the CGS system, it is cm/s.
2. Speed is related to distance while velocity is related to the displacement.
3. If the motion is along a straight line, the values of speed and velocity are the same, otherwise they can be different.

Velocity is the displacement that occurs in unit time.

In the first example (on page 1), the straight line distance between the houses of Sheetal and Sangeeta is 500 m and that between Sangeeta's house and school is 1200 m. Also, the straight line distance between Sheetal's house and school is 1300 m. Suppose Sheetal takes 5 minutes to reach Sangeeta's house and then 24 minutes to reach school from there. Then,

$$\text{Sheetal's speed along path A} = \frac{\text{Distance}}{\text{Time}} = \frac{500 \text{ m}}{5 \text{ minute}} = 100 \text{ m/minute}$$

$$\text{Sheetal's speed along path B} = \frac{\text{Distance}}{\text{Time}} = \frac{1200 \text{ m}}{24 \text{ minute}} = 50 \text{ m/minute}$$

$$\text{Sheetal's average speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{1700 \text{ m}}{29 \text{ minute}} = 58.6 \text{ m/minute}$$

$$\text{Sheetal's velocity} = \frac{\text{Displacement}}{\text{Time}} = \frac{1300 \text{ m}}{29 \text{ minute}}$$

$$\text{Sheetal's velocity} = 44.83 \text{ m/minute}$$

Effect of speed and direction on velocity

Sachin is travelling on a motorbike. Explain what will happen in the following events during Sachin's ride (see figure 1.3).

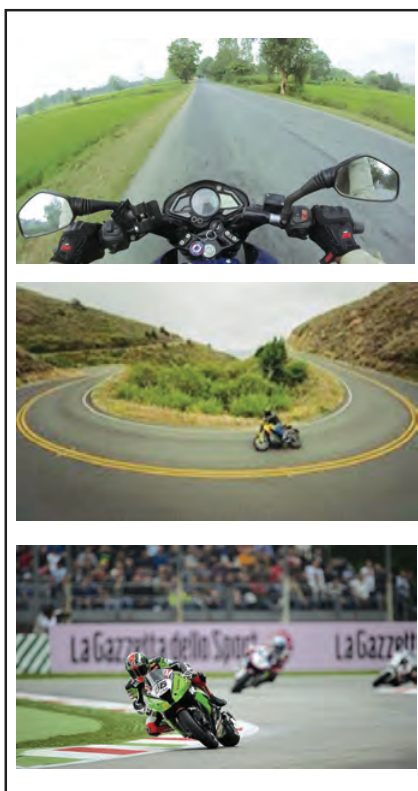
1. What will be the effect on the velocity of the motorcycle if its speed increases or decreases, but its direction remains unchanged?
2. In case of a turning on the road, will the velocity and speed be same?

If Sachin changes the direction of the motorcycle, keeping its speed constant, what will be the effect on the velocity?

3. If, on a turning, Sachin changes the direction as well as the speed of the motorcycle, what will be the effect on its velocity?

It is clear from the above that velocity depends on speed as well as direction and that velocity changes by

1. changing the speed while keeping the direction same
2. changing the direction while keeping the speed same
3. changing the speed as well as the direction.



1.3 Effect on velocity



Always remember

The first scientist to measure speed as the distance /time was Galileo. The speed of sound in dry air is 343.2 m/s while the speed of light is about 3×10^8 m/s. The speed of revolution of the earth around the sun is about 29770 m/s.

Uniform and non-uniform linear motion

Amar, Akbar and Anthony are travelling in different cars with different velocities. The distances covered by them during different time intervals are given in the following table.

Time in the clock	Distance covered by Amar in km	Distance covered by Akbar in km	Distance covered by Anthony in km
5.00	0	0	0
5.30	20	18	14
6.00	40	36	28
6.30	60	42	42
7.00	80	70	56
7.30	100	95	70
8.00	120	120	84



Use your brain power !

1. What is the time interval between the notings of distances made by Amar, Akbar and Anthony?
2. Who has covered equal distances in equal time intervals?
3. Are all the distances covered by Akbar in the fixed time intervals the same?
4. Considering the distances covered by Amar, Akbar and Anthony in fixed time intervals, what can you say about their speeds?

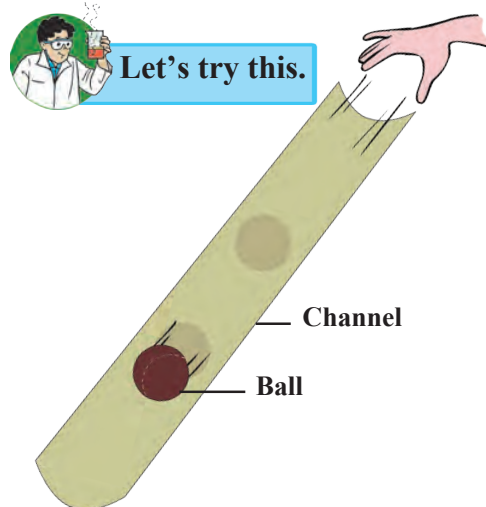
If an object covers equal distances in equal time intervals, it is said to be moving with uniform speed.

If an object covers unequal distances in equal time intervals, it is said to be moving with non-uniform speed. For example, the motion of a vehicle being driven through heavy traffic.

Acceleration



Let's try this.



1.4 Change in velocity

1. Take a 1m long plastic tube and cut it lengthwise into two halves.
2. Take one of the channel shaped pieces. Place one of its ends on the ground and hold the other at some height from the ground as shown in figure 1.4.
3. Take a small ball and release it from the upper end of the channel.
4. Observe the velocity of the ball as it rolls down along the channel.
5. Is its velocity the same at all points?
6. Observe how the velocity changes as it moves from the top, through the middle and to the bottom.

You must have all played on a slide in a park. You know that while sliding down, the velocity is less at the top, it increases in the middle and becomes zero towards the end. The rate of change of velocity is called acceleration

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time.}}$$

If the initial velocity is 'u' and in time 't' it changes to the final velocity 'v',

$$\text{Acceleration} = a = \frac{\text{Final velocity} - \text{Initial velocity}}{\text{Time}} \therefore a = \frac{(v-u)}{t}$$

If the velocity of an object changes during a certain time period, then it is said to have accelerated motion. An object in motion can have two types of acceleration.

1. When an object is at rest in the beginning of its motion, what is its initial velocity?
2. When an object comes to rest at the end of its motion, what is its final velocity?

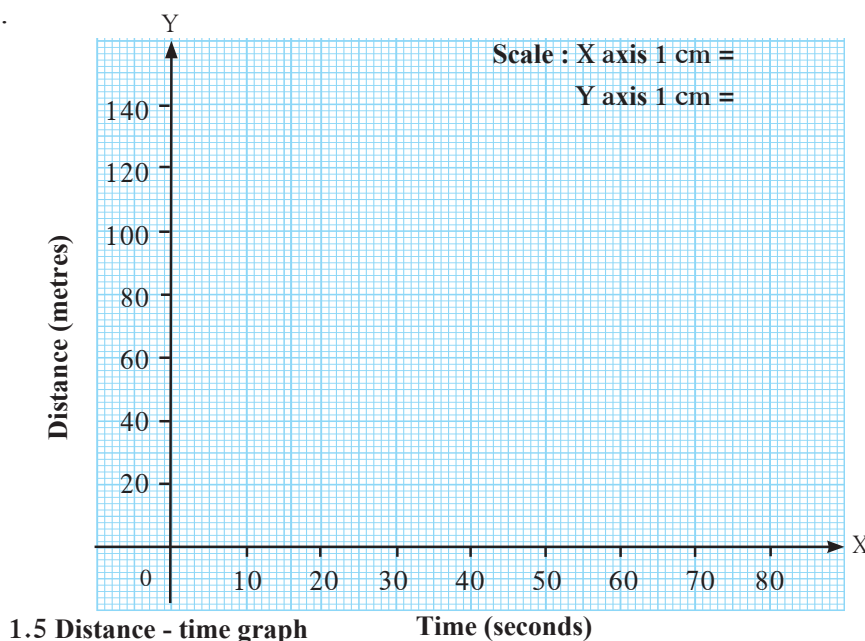
Positive, negative and zero acceleration

An object can have positive or negative acceleration. When the velocity of an object increases, the acceleration is positive. In this case, the acceleration is in the direction of velocity. When the velocity of an object decreases with time, it has negative acceleration. Negative acceleration is also called deceleration. Its direction is opposite to the direction of velocity. If the velocity of the object does not change with time, it has zero acceleration.

Distance-time graph for uniform motion

The following table shows the distances covered by a car in fixed time intervals. Draw a graph of distance against time taking 'time' along the X-axis and 'distance' along the Y-axis in figure 1.5.

Time (sec-onds)	Distance (metres)
0	0
10	15
20	30
30	45
40	60
50	75
60	90
70	105



1.5 Distance - time graph



Use your brain power!

1. If the velocity changes by equal amounts in equal time intervals, the object is said to be in uniform acceleration.
2. If the velocity changes by unequal amounts in equal time intervals, the object is said to be non-uniform acceleration.

An object in uniform motion covers equal distances in equal time intervals. Thus, the graph between distance and time is a straight line.



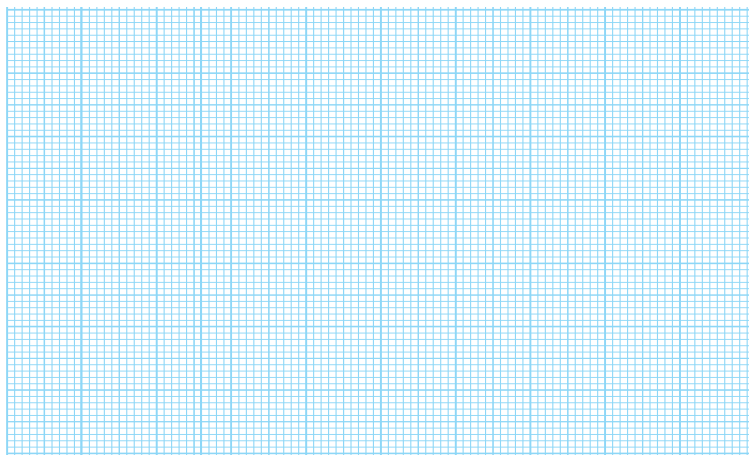
Use your brain power !

In the distance-time graph above, what does the slope of the straight line indicate?

Distance-time graph for non-uniform motion

The following table shows the distances covered by a bus in equal time intervals. Draw a graph of distance against time taking the time along the X-axis and distance along the Y-axis in figure 1.6. Does the graph show a direct proportionality between distance and time?

Time (second)	Distance (metre)
0	0
5	7
10	12
15	20
20	30
25	41
30	50
35	58



1.6 Distance - time graph

Here, the distance changes non-uniformly with time. Thus, the bus is having non-uniform motion.



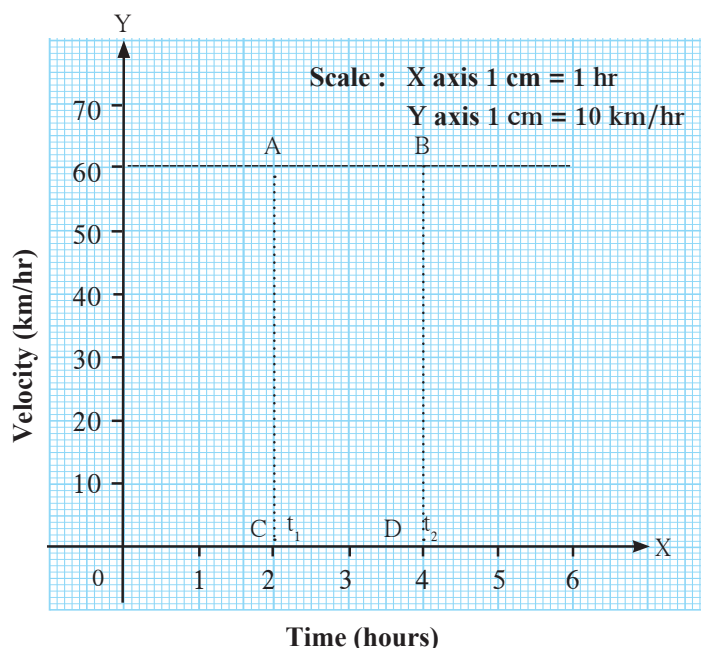
Use your brain power !

What difference do you see in the distance-time graphs for uniform and non-uniform motion?

Velocity-time graph for uniform velocity

A train is moving with a uniform velocity of 60 km/hour for 5 hours. The velocity-time graph for this uniform motion is shown in figure 1.7.

1. With the help of the graph, how will you determine the distance covered by the train between 2 and 4 hours?
2. Is there a relation between the distance covered by the train between 2 and 4 hours and the area of a particular quadrangle in the graph? What is the acceleration of the train?

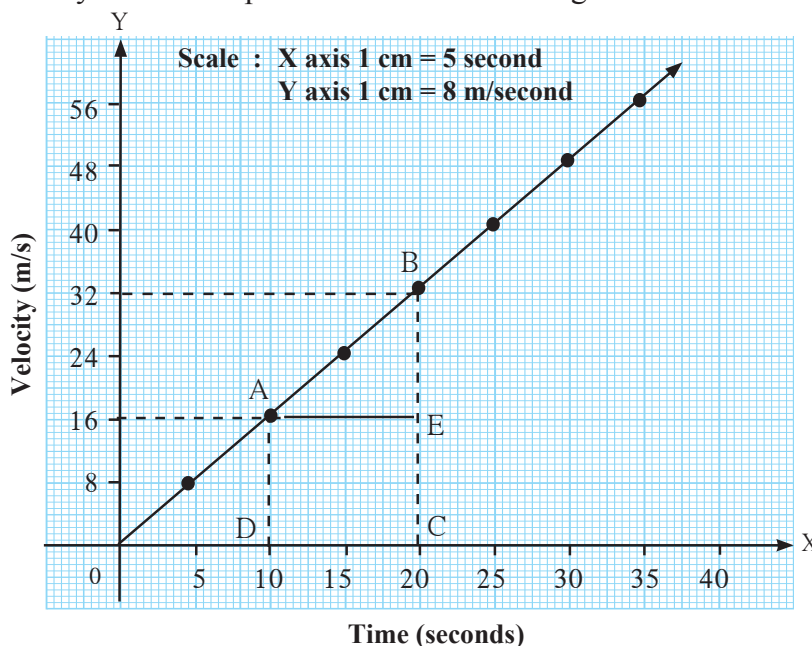


1.7 Velocity - time graph

Velocity-time graph for uniform acceleration

The changes in the velocity of a car in specific time intervals are given in the following table.

Time (seconds)	Velocity (m/s)
0	0
5	8
10	16
15	24
20	32
25	40
30	48
35	56



1.8 Velocity - time graph

The velocity-time graph in figure 1.8 shows that,

1. The velocity changes by equal amounts in equal time intervals. Thus, this is uniform acceleration in accelerated motion. How much does the velocity change every 5 minutes?
2. For all uniformly accelerated motions, the velocity-time graph is a straight line.
3. For non-uniformly accelerated motions, the velocity-time graph may have any shape depending on how the acceleration changes with time.

From the graph in figure 1.8, we can determine the distance covered by the car between the 10th and the 20th seconds as we did in the case of the train in the previous example. The difference is that the velocity of the car is not constant (unlike that of the train) but is continuously changing because of uniform acceleration. In such a case, we have to use the average velocity of the car in the given time interval to determine the distance covered in that interval.

From the graph, the average velocity of the car = $\frac{32+16}{2} = 24 \text{ m/s}$

Multiplying this by the time interval, i.e. 10 seconds gives us the distance covered by the car. Distance covered = $24 \text{ m/s} \times 10 \text{ s} = 240 \text{ m}$

Check that, similar to the example of the train, the distance covered is given by the area of quadrangle ABCD.

$$A(\square ABCD) = A(\square AECD) + A(\triangle ABE)$$

Equations of motion using graphical method

Newton studied motion of an object and gave a set of three equations of motion. These relate the displacement, velocity, acceleration and time of an object moving along a straight line.

Suppose an object is in motion along a straight line with initial velocity 'u'. It attains a final velocity 'v' in time 't' due to acceleration 'a' its displacement is 's'. The three equations of motion can be written as

$v = u + at$ This is the relation between velocity and time.

$s = ut + \frac{1}{2} at^2$ This is the relation between displacement and time.

$v^2 = u^2 + 2as$ This is the relation between displacement and velocity.

Let us try to obtain these equations by the graphical method.

Equation describing the relation between velocity and time

Figure 1.9 shows the change in velocity with time of a uniformly accelerated object. The object starts from the point D in the graph with velocity u. Its velocity keeps increasing and after time t, it reaches the point B on the graph.

The initial velocity of the object = $u = OD$

The final velocity of the object = $v = OC$

Time = $t = OE$

$$\begin{aligned} \text{Acceleration (a)} &= \frac{\text{Change in velocity}}{\text{Time}} \\ &= \frac{(\text{Final velocity} - \text{Initial velocity})}{\text{Time}} \\ &= \frac{(OC - OD)}{t} \end{aligned}$$

$$\therefore CD = at \quad \dots\dots (i) \quad (OC - OD = CD)$$

Draw a line parallel to Y axis passing through B. This will cross the X axis in E. Draw a line parallel to X-axis passing through D. This will cross the line BE at A.

From the graph.... $BE = AB + AE$

$$\therefore v = CD + OD \quad \dots\dots (AB = CD \text{ and } AE = OD)$$

$$\therefore v = at + u \quad \dots\dots\dots (\text{from } i)$$

$$\therefore v = u + at$$

This is the first equation of motion.

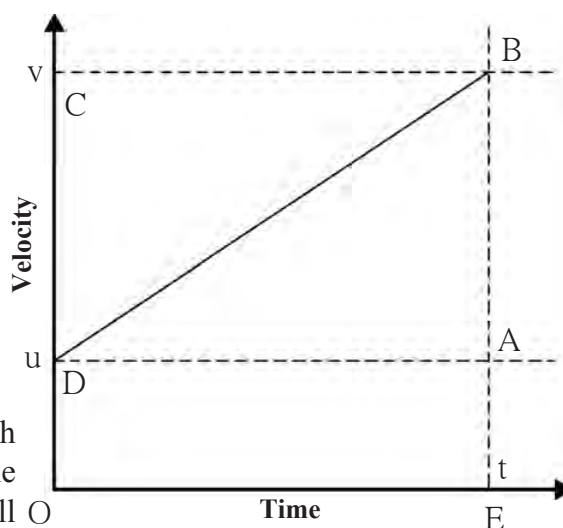
Equation describing the relation between displacement and time

Let us suppose that an object in uniform acceleration 'a' and it has covered the distance 's' within time 't'. From the graph in figure 1.9, the distance covered by the object during time 't' is given by the area of quadrangle DOEB.

$$\therefore s = \text{area of quadrangle DOEB}$$

$$= \text{area (rectangle DOEA)} + \text{area of triangle (DAB)}$$

$$\therefore s = (AE \times OE) + \left(\frac{1}{2} \times [AB \times DA] \right)$$



1.9 Velocity - time graph

But, $AE = u$, $OE = t$ and $(OE = DA = t)$
 $AB = at$ --- ($AB = CD$) --- from (i)

$$\therefore s = u \times t + \frac{1}{2} \times at \times t$$

$$\therefore \text{Newton's second equation of motion is } s = ut + \frac{1}{2} at^2$$

Equation describing the relation between displacement and velocity

We have seen that from the graph in figure 1.9 we can determine the distance covered by the object in time t from the area of the quadrangle DOEB. DOEB is a trapezium. So we can use the formula for its area.

$\therefore s = \text{area of trapezium DOEB}$

$$\therefore s = \frac{1}{2} \times \text{sum of lengths of parallel sides} \times \text{distance between the parallel sides}$$

$$\therefore s = \frac{1}{2} \times (OD + BE) \times OE \quad \text{But, } OD = u, BE = v \text{ and } OE = t$$

$$\therefore s = \frac{1}{2} \times (u + v) \times t \quad \text{----- (ii)}$$

$$\text{But, } a = \frac{(v-u)}{t}$$

$$\therefore t = \frac{(v-u)}{a} \quad \text{----- (iii)}$$

$$\therefore s = \frac{1}{2} \times (u + v) \times \frac{(v-u)}{a}$$

$$\therefore s = \frac{(v+u)(v-u)}{2a}$$

$$\therefore 2as = (v+u)(v-u) = v^2 - u^2$$

$$\therefore v^2 = u^2 + 2as$$

this is Newton's third equation of motion.



Always remember

The velocity of an accelerated object changes with time. Change in the velocity can be due to a change in direction or magnitude of the velocity or both.

Uniform circular motion



Let's try this

Observe the tip of the second hand of a clock. What can you say about its velocity and speed?

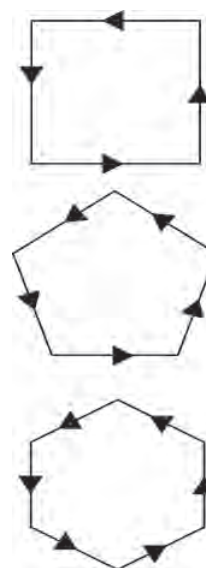
The speed of the tip of a clock is constant, but the direction of its displacement and therefore, its velocity is constantly changing. As the tip is moving along a circular path, its motion is called uniform circular motion. Can you give more examples of such motion?



Try out and think about it

1. Draw a rectangular path as shown figure 1.10
2. Place the tip of your pencil on the middle of any side of the square path and trace the path.
3. Note how many times you have to change the direction while tracing the complete path.
4. Now repeat this action for a pentagonal, hexagonal, octagonal path and note the number of times you have to change direction.
5. If you increase the number of sides of the polygon and make it infinite, how many times will you have to change the direction? What will be the shape of the path?

This shows that as we increase the number of sides, we have to keep changing direction more and more times. And when we increase the number of sides to infinity, the polygon becomes a circle.



1.10 Changes in direction

When an object is moving with a constant speed along a circular path, the change in velocity is only due to the change in direction. Hence, it is accelerated motion. When an object moves with constant speed along a circular path, the motion is called uniform circular motion, e.g. the motion of a stone in a sling or that of any point on a bicycle wheel when they are in uniform motion.

If an object, moving along a circular path of radius 'r', takes time 't' to come back to its starting position, its speed can be determined using the formula given below :

$$\text{Speed} = \frac{\text{Circumference}}{\text{Time}}$$

$$v = \frac{2 \pi r}{t} \quad r = \text{radius of the circle}$$



Research

Find out more examples of circular motion in day to day life.

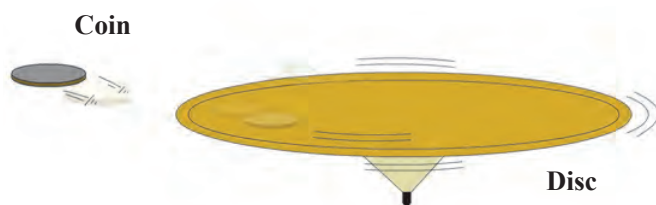
Determining the direction of velocity in uniform circular motion.



Let's try this

Take a circular disc and put a five rupee coin at a point along its edge.

Make it move around its axis by putting a pin through it. When the disc is moved at higher speed, the coin will be thrown off as shown in figure 1.11. Note the direction in which it is thrown off. Repeat the action placing the coin at different points along the edge of the circle and observe the direction in which the coin is thrown off.



1.11 The coin and the disc

The coin will be thrown off in the direction of the tangent which is perpendicular to the radius of disc. Thus, the direction in which it gets thrown off depends on its position at the moment of getting thrown off. It means that, as the coin moves along a circular path the direction of its motion is changing at every point.

Solved examples

Example 1: An athlete is running on a circular track. He runs a distance of 400 m in 25 s before returning to his original position. What is his average speed and velocity?

Given : Total distance travelled = 400 m

Total displacement = 0, as he returns to his original position.

Total time = 25 seconds.

Average speed = ?, Average velocity = ?

$$\text{Average speed} = \frac{\text{Total distance covered}}{\text{Total time taken}} = \frac{400}{25} = 16 \text{ m/s}$$

$$\text{Average velocity} = \frac{\text{Total displacement}}{\text{Total time taken}} = \frac{0}{25} = 0 \text{ m/s}$$

Example 2: An aeroplane taxis on the runway for 30 s with an acceleration of 3.2 m/s^2 before taking off. How much distance would it have covered on the runway?

Given : $a = 3.2 \text{ m/s}^2$, $t = 30 \text{ s}$, $u = 0$, $s = ?$

$$s = ut + \frac{1}{2}at^2 = 0 \times 30 + \frac{1}{2} \times 3.2 \times 30^2 = 1440 \text{ m.}$$

Example 3: A kangaroo can jump 2.5 m vertically. What must be the initial velocity of the kangaroo?

Given :

$$a = 9.8 \text{ m/s}^2$$

$$s = 2.5 \text{ m}$$

$$v = 0$$

$$u = ?$$

$$v^2 = u^2 + 2as$$

$(0)^2 = u^2 + 2 \times (-9.8) (2.5)$ Negative sign is used as the acceleration is in the direction opposite to that of velocity.

$$0 = u^2 - 49$$

$$u^2 = 49$$

$$u = 7 \text{ m/s}$$

Example 4 : A motorboat starts from rest and moves with uniform acceleration. If it attains the velocity of 15 m/s in 5 s , calculate the acceleration and the distance travelled in that time.

Given :

Initial velocity, $u = 0$,

final velocity, $v = 15 \text{ m/s}$, time, $t = 5 \text{ s}$.

Acceleration = $a = ?$

From the first equation of motion

$$a = \frac{v-u}{t} = \frac{15-0}{5} = 3 \text{ m/s}^2$$

From the second equation of motion, the distance covered will be

$$\begin{aligned}s &= ut + \frac{1}{2} at^2 \\s &= 0 \times 5 + \frac{1}{2} 3 \times 5^2 \\&= 0 + \frac{75}{2} = 37.5 \text{ m}\end{aligned}$$

Newton's laws of motion

What could be the reason for the following?

1. A static object does not move without the application of a force.
2. The force which is sufficient to lift a book from a table is not sufficient to lift the table.
3. Fruits on a tree fall down when its branches are shaken.
4. An electric fan keeps on rotating for some time even after it is switched off.

If we look for reasons for the above, we realize that objects have some inertia. We have learnt that inertia is related to the mass of the object. Newton's first law of motion describes this very property and is therefore also called the law of inertia.

Newton's first law of motion



Let's try this

Fill a glass with sand. Keep a piece of cardboard on it. Keep a five rupee coin on the cardboard. Now strike the cardboard hard using your fingers. Observe what happens.

Balanced and unbalanced force

You must have played tug-of-war. So long as the forces applied by both the sides are equal, i.e. balanced, the centre of the rope is static in spite of the applied forces. On the other hand, when the applied forces become unequal, i.e. unbalanced, a net force gets applied in the direction of the greater force and the centre of the rope shifts in that direction.

'An object continues to remain at rest or in a state of uniform motion along a straight line unless an external unbalanced force acts on it.'

When an object is at rest or in uniform motion along a straight line, it does not mean that no force is acting on it. Actually there are a number of forces acting on it, but they cancel one another so that the net force is zero. Newton's first law explains the phenomenon of inertia, i.e. the inability of an object to change its state of motion on its own. It also explains the unbalanced forces which cause a change in the state of an object at rest or in uniform motion.

All instances of inertia are examples of Newton's first law of Motion.

Newton's second law of motion



Let's try this

- A.** 1. Ask your friend to drop one plastic and one rubber ball from the same height.
2. You catch the balls. Which ball was easier to catch and why?
- B.** 1. Ask your friend to throw a ball towards you at slow speed. Try to catch it.
2. Now ask your friend to throw the same ball at high speed towards you. Try to catch it. Which ball could you catch with greater ease? Why?

The effect of one object striking another object depends both on the mass of the former object and its velocity. This means that the effect of the force depends on a property related to both mass and velocity of the striking object. This property was termed 'momentum' by Newton.

Momentum has magnitude as well as direction. Its direction is the same as that of velocity. In SI system, the unit of momentum is kg m/s, while in CGS system, it is g cm/s.

If an unbalanced force applied on an object causes a change in the velocity of the object, then it also causes a change in its momentum. The force necessary to cause a change in the momentum of an object depends upon the rate of change of momentum.

Momentum (P) : Momentum is the product of mass and velocity of an object. $P = m v$. Momentum is a vector quantity.

'The rate of change of momentum is proportional to the applied force and the change of momentum occurs in the direction of the force.'

Suppose an object of mass m has an initial velocity u . When a force F is applied in the direction of its velocity for time t , its velocity becomes v .

$\therefore$ The initial momentum of the object = mu ,

Its final momentum after time t = mv

$$\therefore \text{Rate of change of momentum} = \frac{\text{Change in momentum}}{\text{Time}}$$

$$\therefore \text{Rate of change of momentum} = \frac{mv - mu}{t} = \frac{m(v - u)}{t} = ma$$

According to Newton's second law of motion, the rate of change of momentum is proportional to the applied force.

$$\therefore ma \propto F$$

$$\therefore F = k ma \quad (k = \text{Constant of proportionality and its value is } 1).$$

$$F = m \times a$$

Consider two objects having different masses which are initially at rest. The initial momentum for both is zero. Suppose a force 'F' acts for time 't' on both objects. The lighter object starts moving faster than the heavier object. However, from the above formula, we know that the rate of change of momentum i.e. 'F' in both objects is same and the total change in their momentum will also be same i.e. 'Ft'. Thus, if the same force is applied on different objects, the change in momentum is the same.

In SI system, the unit of force is newton.

Newton (N) : The force necessary to cause an acceleration of 1 m/s^2 in an object of mass 1 kg is called 1 newton.

$$1 \text{ N} = 1 \text{ kg} \times 1 \text{ m/s}^2$$

In CGS system the unit of force is a dyne.

Dyne: The force necessary to cause an acceleration of 1 cm/s^2 in an object of mass 1 gm is called 1 dyne.

$$1 \text{ dyne} = 1 \text{ g} \times 1 \text{ cm/s}^2$$



Use your brain power !

Why is there a thick bed of sand for a high jumper to fall on after his jump?

Newton's third law of motion



Let's try this

1. Take a plastic boat and make a hole at its rear end.
2. Inflate a balloon and fix it on the hole in the boat. Release the boat in water. What happens to the boat as the air in the balloon escapes slowly? Why?

We have learnt about force and its effect on an object through Newton's first and second laws of motion.

'However, in nature force cannot act alone.' Force is a reciprocal action between two objects. Forces are always applied in pairs. When one object applies a force on another object, the latter object also simultaneously applies a force on the former object. The forces between two objects are always equal and opposite. This idea is expressed in Newton's third law of motion. The force applied by the first object is called action force while the force applied by the second object on the first is called reaction force.

'Every action force has an equal and opposite reaction force which acts simultaneously.'



Use your brain power !

1. Action and reaction are terms that express force.
2. These forces act in pairs. One force cannot exist by itself.
3. Action and reaction forces act simultaneously.
4. Action and reaction forces act on different objects. They do not act on the same object and hence cannot cancel each other's effect.

1. While hitting a ball with a bat, the speed of the bat decreases.
2. A gun recoils i.e. moves backwards when a bullet is fired.
3. Mechanism of firing of a rocket.

How will you explain these with the help of Newton's third law of motion?

Law of conservation of momentum

Suppose an object A has mass m_1 and its initial velocity is u_1 . An object B has mass m_2 and initial velocity u_2 .

According to the formula for momentum, the initial momentum of A is $m_1 u_1$ and that of B is $m_2 u_2$.

Suppose these two objects collide. Let the force on A due to B be F_1 . This force will cause acceleration in A and its velocity will become v_1 .

$\therefore$ Momentum of A after collision = $m_1 v_1$

According to Newton's third law of motion, A also exerts an equal force on B but in the opposite direction. This will cause a change in the momentum of B. If its velocity after collision is v_2 ,

The momentum of B after collision = $m_2 v_2$. If F_2 is the force that acts on object B,

$$F_2 = -F_1$$

$$\therefore m_2 a_2 = -m_1 a_1 \dots \because F = ma$$

$$\therefore m_2 \times \frac{(v_2 - u_2)}{t} = -m_1 \times \frac{(v_1 - u_1)}{t} \dots \because a = \frac{(v - u)}{t}$$

$$\therefore m_2 (v_2 - u_2) = -m_1 (v_1 - u_1)$$

$$\therefore m_2 v_2 - m_2 u_2 = -m_1 v_1 + m_1 u_1$$

$$\therefore (m_2 v_2 + m_1 v_1) = (m_1 u_1 + m_2 u_2)$$

The magnitude of total final momentum = the magnitude of total initial momentum.

Thus, if no external force is acting on two objects, then their total initial momentum is equal to their total final momentum. This statement is true for any number of objects.

‘When no external force acts on two interacting objects, their total momentum remains constant. It does not change.’

This is a corollary to Newton's third law of motion. The momentum is unchanged after the collision. The momentum gets redistributed between the colliding objects. The momentum of one of the objects decreases while that of the other increases. Thus, we can also state this corollary as follows.

‘When two objects collide, the total momentum before collision is equal to the total momentum after collision.’

In order to understand this principle, let us consider the example of a bullet fired from a gun. When a bullet of mass m_1 is fired from a gun of mass m_2 , its velocity becomes v_1 , and its momentum becomes $m_1 v_1$. Before firing the bullet, both the gun and the bullet are at rest and hence the total initial momentum is zero. According to the above law, the total final momentum also has to be zero. Thus, the forward moving bullet causes the gun to move backward after firing. This backward motion of the gun is called its **recoil**. The velocity of recoil, v_2 , is such that,

$$m_1 v_1 + m_2 v_2 = 0 \text{ or } v_2 = - \frac{m_1}{m_2} \times v_1$$

As the mass of the gun is much higher than the mass of the bullet, the velocity of the gun is much smaller than the velocity of the bullet. The magnitude of the momentum of the bullet and that of the gun are equal and their directions are opposite. Thus, the total momentum is constant. Total momentum is also constant during the launch of a rocket.

Solved examples

Example 1: The mass of a cannon is 500 kg and it recoils with a speed of 0.25 m/s. What is the momentum of the cannon?

Given : mass of the cannon = 500 kg, recoil speed = 0.25 m/s Momentum = ?

Momentum = $m \times v = 500 \times 0.25 = 125 \text{ kg m/s}$

Example 2: 2 balls have masses of 50 gm and 100 gm respectively and they are moving along the same line in the same direction with velocities of 3 m/s and 1.5 m/s respectively. They collide with each other and after the collision, the first ball moves with a velocity of 2.5 m/s. Calculate the velocity of the other ball after collision.

Given :

The mass of first ball = $m_1 = 50 \text{ g} = 0.05 \text{ kg}$, mass of the second ball = $m_2 = 100 \text{ g} = 0.1 \text{ kg}$

Initial velocity of the first ball = $u_1 = 3 \text{ m/s}$, Initial velocity of the second ball = $u_2 = 1.5 \text{ m/s}$

Final velocity of the first ball = $v_1 = 2.5 \text{ m/s}$, Final velocity of the second ball = $v_2 = ?$

According to the law of conservation of momentum, total initial momentum = Total final momentum.

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$(0.05 \times 3) + (0.1 \times 1.5) = (0.05 \times 2.5) + (0.1 \times v_2)$$

$$\therefore (0.15) + (0.15) = 0.125 + 0.1 v_2$$

$$\therefore 0.3 = 0.125 + 0.1 v_2$$

$$\therefore 0.1 v_2 = 0.3 - 0.125 \quad \therefore v_2 = \frac{0.175}{0.1} = 1.75 \text{ m/s}$$



Exercises

1. Match the first column with appropriate entries in the second and third columns and remake the table.

S. No.	Column 1	Column 2	Column 3
1	Negative acceleration	The velocity of the object remains constant	A car, initially at rest reaches a velocity of 50 km/hr in 10 seconds
2	Positive acceleration	The velocity of the object decreases	A vehicle is moving with a velocity of 25 m/s
3	Zero acceleration	The velocity of the object increases	A vehicle moving with the velocity of 10 m/s, stops after 5 seconds.

2. Clarify the differences

- Distance and displacement
- Uniform and non-uniform motion.

3. Complete the following table.

u (m/s)	a (m/s ²)	t (sec)	v = u + at (m/s)
2	4	3	–
–	5	2	20

u (m/s)	a (m/s ²)	t (sec)	s = ut + $\frac{1}{2}$ at ² (m)
5	12	3	–
7	–	4	92

u (m/s)	a (m/s ²)	s (m)	v ² = u ² + 2as (m/s) ²
4	3	–	8
–	5	8.4	10

4. Complete the sentences and explain them.

- The minimum distance between the start and finish points of the motion of an object is called the of the object.
- Deceleration is ----- acceleration
- When an object is in uniform circular motion, its changes at every point.
- During collision remains constant.
- The working of a rocket depends on Newton's law of motion.

5. Give scientific reasons.

- When an object falls freely to the ground, its acceleration is uniform.
- Even though the magnitudes of action force and reaction force are equal and their directions are opposite, their effects do not get cancelled.
- It is easier to stop a tennis ball as compared to a cricket ball, when both are travelling with the same velocity.
- The velocity of an object at rest is considered to be uniform.

6. Take 5 examples from your surroundings and give explanation based on Newtons laws of motion.

7. Solve the following examples.

- An object moves 18 m in the first 3 s, 22 m in the next 3 s and 14 m in the last 3 s. What is its average speed? **(Ans: 6 m/s)**
- An object of mass 16 kg is moving with an acceleration of 3 m/s². Calculate the applied force. If the same force is applied on an object of mass 24 kg, how much will be the acceleration? **(Ans: 48 N, 2 m/s²)**
- A bullet having a mass of 10 g and moving with a speed of 1.5 m/s, penetrates a thick wooden plank of mass 900 g. The plank was initially at rest. The bullet gets embedded in the plank and both move together. Determine their velocity. **(Ans: 0.15 m/s)**
- A person swims 100 m in the first 40 s, 80 m in the next 40 s and 45 m in the last 20 s. What is the average speed? **(Ans: 2.25 m/s²)**

Project:

Obtain information about commonly used gadgets or devices which are based on the principles of Newton's laws of motion.



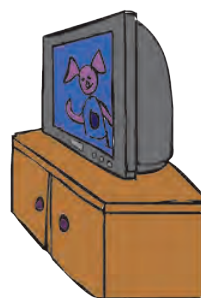
2. Work and Energy



- Work
- Energy
- Mechanical energy
- Law of conservation of energy
- Free fall



Observe



2.1 Various activities



Can you tell?

1. In which of the pictures above has work been done?
2. From scientific point of view, when do we say that no work was done?

Generally, any mental or physical activity is referred to as work. When we walk or run, the energy in our body is used to do the necessary work. We say that a girl who is studying is working or performing work. But that is mental work.

In physics, we deal with physical work. Work has a special meaning in physics.

‘Work is said to be done when a force applied on an object causes displacement of the object.’

You have already learnt that the work done by a force acting on an object is the product of the magnitude of the force and the displacement of the object in the direction of the force. Thus, $\text{Work} = \text{force} \times \text{displacement}$



Can you recall?

What are different types of forces and their examples?

Minakshee wants to displace a wooden block from point A to point B along the surface of a table as shown in figure 2.2A. She has used force F for the purpose. Has all the energy she spent been used to produce acceleration in the block? Which forces have been overcome using that energy?



Use your brain power !

You have learnt how to calculate the work done on an object when the displacement is in the direction of the applied force. But if the displacement is not in the direction of the applied force, how do we calculate the amount of work done?

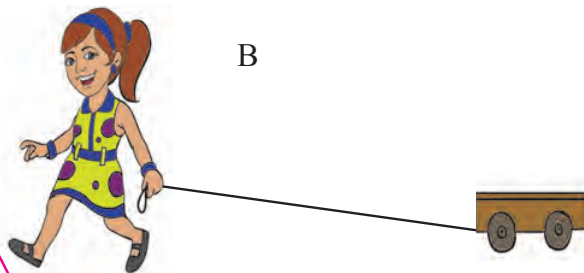


Observe and discuss.

A



B



C



2.2 Displacement of an object

You must have seen the events depicted in the pictures B and C above. When a child pulls a toy with the help of a string, the direction of the force is different from that of displacement. Similarly, when a large vehicle tows a small one, the directions of force and the displacements are different. In both these cases, the direction of force makes an angle with the direction of displacement. Let us see how to calculate work done in such cases.

When a child pulls a toy cart, force is applied along the direction of the string while the cart is pulled along the horizontal surface. In this case, in order to calculate the amount of work done, we have to convert the applied force into the force acting along the direction of displacement.

Let F be the applied force and F_1 be its component in the direction of displacement. Let s be the displacement. The amount of work done is given by

$$W = F_1 \cdot s \quad \dots\dots\dots (1)$$

The force F is applied in the direction of the string i. e. at an angle with the horizontal. Let θ be the angle that the string makes with the horizontal. We can determine the component F_1 , of this force F , which acts in the horizontal direction by means of trigonometry.

(see figure 2.3)

$$\cos \theta = \text{base} / \text{hypotenuse}$$

$$\cos \theta = F_1 / F$$

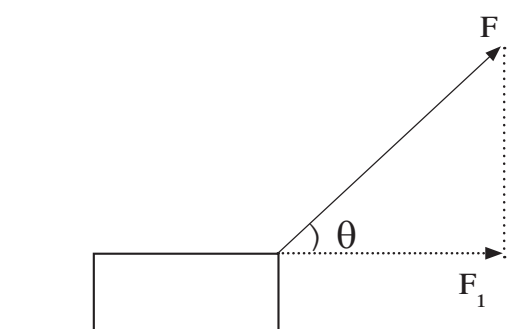
$$F_1 = F \cos \theta$$

Thus, the work done by F_1 is

$$W = F \cos \theta \cdot s$$

$$W = F s \cos \theta$$

Enter your conclusions about the work done for the specific values of θ in the following table.



2.3 Force used for the displacement

θ	$\cos \theta$	$W = F s \cos \theta$	Conclusion
0°	1	$W = F s$	
90°	0	0	
180°	-1	$W = -F s$	

Unit of work

Work = Force $\times$ Displacement

In SI system, the unit of force is newton (N) and the unit of displacement is metre (m). Thus, the unit of force is newton-metre. This is called joule.

1 Joule : If a force of 1 newton displaces an object through 1 metre in the direction of the force, the amount of work done on the object is 1 joule.

$$\therefore 1 \text{ joule} = 1 \text{ newton} \times 1 \text{ metre}$$

$$1 \text{ J} = 1 \text{ N} \times 1 \text{ m}$$

In CGS system, the unit of force is dyne and that of displacement is centimeter (cm). Thus, the unit of work done is dyne-centimetre. This is called an erg.

1 erg : If a force of 1 dyne displaces an object through 1 centimetre in the direction of the force, the amount of work done is 1 erg.

$$1 \text{ erg} = 1 \text{ dyne} \times 1 \text{ cm}$$

Relationship between joule and erg

We know that, 1 newton = 10^5 dyne and 1 m = 10^2 cm

Work = force $\times$ displacement

$$1 \text{ joule} = 1 \text{ newton} \times 1 \text{ m}$$

$$1 \text{ joule} = 10^5 \text{ dyne} \times 10^2 \text{ cm}$$

$$= 10^7 \text{ dyne cm}$$

$$1 \text{ joule} = 10^7 \text{ erg}$$

Positive, negative and zero work



Think before you answer

Discuss the directions of force and of displacement in each of the following cases.

1. Pushing a stalled vehicle
2. Catching the ball which your friend has thrown towards you.
3. Tying a stone to one end of a string and swinging it round and round by the other end of the string.
4. Walking up and down a staircase; climbing a tree.
5. Stopping a moving car by applying brakes.

You will notice that in some of the above examples, the direction of the force and displacement are the same. In some other cases, these directions are opposite to each other, while in some cases, they are perpendicular to each other. In these cases, the work done by the force is as follows.

1. When the force and the displacement are in the same direction ($\theta = 0^\circ$), the work done by the force is positive.
2. When the force and the displacement are in opposite directions ($\theta = 180^\circ$), the work done by the force is negative.
3. When the applied force does not cause any displacement or when the force and the displacement are perpendicular to each other ($\theta = 90^\circ$), the work done by the force is zero.



Try this

Take a plastic cup and make a hole in the centre of its bottom. Take a long thread, double it and pass it through the hole. Tie a thick enough knot at the end so that the knot will not pass through the hole, taking care that the two loose ends are below the bottom of the cup. Tie a nut each to the two ends as shown in figure 2.4. Now do the following.

As shown in figure 'A', put the cup on a table, keep one of the nuts in the cup and let the thread carrying the other nut hang down along the side of the table. What happens?

As shown in figure 'B', when the cup is sliding along the table, stop it by putting a ruler in its path.

As shown in figure 'C', keep the cup at the centre of the table and leave the two nuts hanging on opposite sides of the table.

Questions:

1. Figure A- Why does the cup get pulled?
2. Figure B- What is the relation between the displacement of the cup and the force applied through the ruler?
3. In figure C- Why doesn't the cup get displaced?
4. What is the type of work done in figures A, B and C?

In the three actions above, what is the relationship between the applied force and the displacement?

Figure A

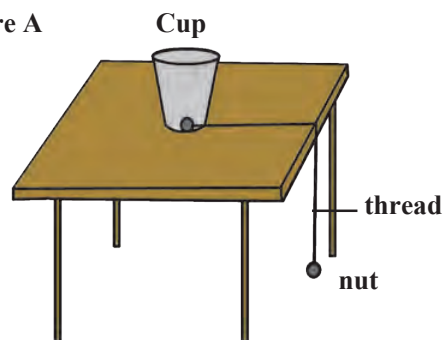


Figure B

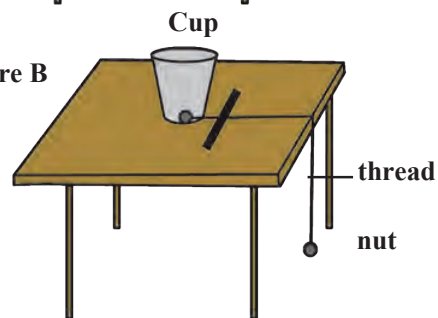
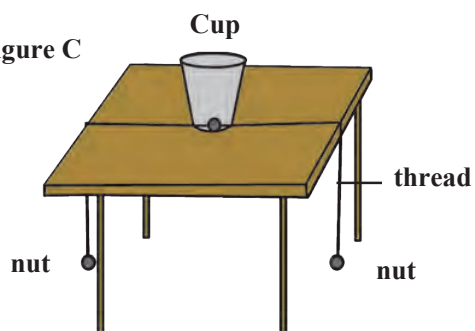


Figure C



2.4 Positive, negative and zero work

Suppose an artificial satellite is moving around the earth in a circular orbit. As the gravitational force acting on the satellite (along the radius of the circle) and its displacement (along the tangent to the circle) are perpendicular to each other, the work done by the gravitational force is zero

Institutes at work

The National Physical Laboratory, New Delhi, was conceptualized in 1943. It functions under the Council of Scientific and Industrial Research. It conducts basic research in the various branches of physics and helps various industries and institutes engaged in developmental work. Its main objective is to establish national standards of various physical quantities.

Solved examples

Example 1: Calculate the work done to take an object of mass 20 kg to a height of 10 m.

$$(g = 9.8 \text{ m/s}^2)$$

Given: $m = 20 \text{ kg}$; $s = 10 \text{ m}$

$$g = 9.8 \text{ m/s}^2$$

$$\therefore F = m \cdot g$$

$$= 20 \times (-9.8)$$

(The negative sign is taken because the displacement is opposite to the direction of the force.)

$$F = -196 \text{ N}$$

$$\therefore W = F s$$

$$= -196 \times 10$$

$$W = -1960 \text{ J}$$

(The negative sign appears because the direction of force is opposite to the direction of displacement so that the work done is negative.)

Example 2 : Pravin has applied a force of 100 N on an object, at an angle of 60° to the horizontal. The object gets displaced in the horizontal direction and 400 J work is done. What is the displacement of the object?

$$(\cos 60^\circ = \frac{1}{2})$$

Given :

$$\theta = 60^\circ$$

$$F = 100 \text{ N}$$

$$W = 400 \text{ J}, s = ?$$

$$W = F s \cos \theta$$

$$400 = 100 \times s \times \frac{1}{2}$$

$$\frac{400}{100} = \frac{1}{2} \times s$$

$$4 \times 2 = s$$

$$\therefore s = 8 \text{ m}$$

The object will be displaced through 8 m.

Energy

Why does it happen?

1. If a pot having a plant is kept in the dark, the plant languishes.
2. On increasing the volume of a music system or TV beyond a limit, the vessels in the house start vibrating.
3. Collecting sunlight on a paper with the help of a convex lens burns the paper.

The capacity of a body to perform work is called its energy. The units of work and energy are the same. The unit in SI system is **joule** while that in cgs system is **erg**.

You have learnt that energy exists in various forms like mechanical, heat, light, sound, electro-magnetic, chemical, nuclear and solar. In this chapter, we are going to study two forms of mechanical energy, namely, potential energy and kinetic energy.

Kinetic energy

What will happen in the following cases?

1. A fast cricket ball strikes the stumps.
2. The striker hits a coin on the carom board.
3. One marble strikes another in a game of marbles.

From the above examples we understand that when a moving object strikes a stationary object, the stationary object moves. Thus, the moving object has some energy, part or all of which it shares with the stationary object, thereby setting it in motion. 'The energy which an object has because of its motion is called its kinetic energy'. The work done by a force to displace a stationary object through a distance s is the kinetic energy gained by the object.

$$\text{Kinetic energy} = \text{work done on the object}$$

$$\therefore \text{K.E.} = F \times s$$

Expression for kinetic energy :

Suppose a stationary object of mass m moves because of an applied force. Let u be its initial velocity (here $u = 0$). Let the applied force be F . This generates an acceleration a in the object, and, after time t , the velocity of the object becomes equal to v . The displacement during this time is s . The work done on the object, $W = F \cdot s$

$$W = F \times s$$

According to Newton's second law of motion,

$$F = ma \text{ ----- (1) Similarly, using Newton's second equation of motion}$$

$$s = ut + \frac{1}{2} at^2 \quad \text{However, as initial velocity is zero, } u = 0.$$

$$s = 0 + \frac{1}{2} at^2$$

$$s = \frac{1}{2} at^2 \text{ -----(2)}$$

$$\therefore W = ma \times \frac{1}{2} at^2 \text{ ----- using equations (1) and (2)}$$

$$W = \frac{1}{2} m(at)^2 \text{ -----(3)}$$

Using Newton's first equation of motion

$$v = u + at$$

$$\therefore v = 0 + at$$

$$\therefore v = at$$

$$\therefore v^2 = (at)^2 \text{ -----(4)}$$

$$\therefore W = \frac{1}{2} mv^2 \text{ ----- using equations (3) and (4)}$$

The kinetic energy gained by an object is the amount of work done on the object.

$$\therefore \text{K. E.} = W$$

$$\therefore \text{K. E.} = \frac{1}{2} mv^2$$

Example : A stone having a mass of 250 gm is falling from a height. How much kinetic energy does it have at the moment when its velocity is 2 m/s?

Given : $m = 250 \text{ g}$ $m = 0.25 \text{ kg}$

$$v = 2 \text{ m/s}$$

$$\text{K.E.} = \frac{1}{2} mv^2 = \frac{1}{2} \times 0.25 \times (2)^2 = 0.5 \text{ J}$$



Use your brain power !

The mass of a moving body is doubled, how many times will the kinetic energy increase?

Potential energy



Try this

1. An arrow is released from a stretched bow.
2. Water kept at a height flows through a pipe into the tap below.
3. A compressed spring is released.

Which words describe the state of the object in the above examples? Where did the energy required to cause the motion of objects come from?

If the objects were not brought in those states, would they have moved?

‘The energy stored in an object because of its specific state or position is called its potential energy.’

1. Hold a chalk at a height of 5 cm from the floor and release it.
2. Now stand up straight and then release the chalk.
3. Is there a difference in the results of the two activities? If so, why?

Expression for potential energy

To carry an object of mass ‘m’ to a height ‘h’ above the earth’s surface, a force equal to ‘mg’ has to be used against the direction of the gravitational force. The amount of work done can be calculated as follows.

Work = force x displacement

$$W = mg \times h$$

$$\therefore W = mgh$$

$\therefore$ The amount of potential energy stored in the object because of its displacement

$$\text{P.E.} = mgh \quad (W = \text{P.E.})$$

$\therefore$ Displacement to height h causes energy equal to mgh to be stored in the object.

Example : 500 kg water is stored in the overhead tank of a 10 m high building. Calculate the amount of potential energy stored in the water.

Given :

$$h = 10 \text{ m}, m = 500 \text{ kg} \quad g = 9.8 \text{ m/s}^2$$

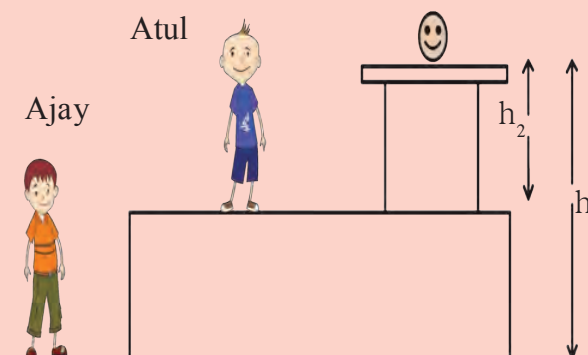
$$\therefore \text{P.E.} = mgh$$

$$= 10 \times 9.8 \times 500$$

$$\text{P.E.} = 49000 \text{ J}$$

Ajay and Atul have been asked to determine the potential energy of a ball of mass m kept on a table as shown in the figure. What answers will they get? Will they be different? What do you conclude from this?

Potential energy is relative. The heights of the ball with respect to Ajay and Atul are different. So the potential energy with respect to them will be different.



Transformation of energy

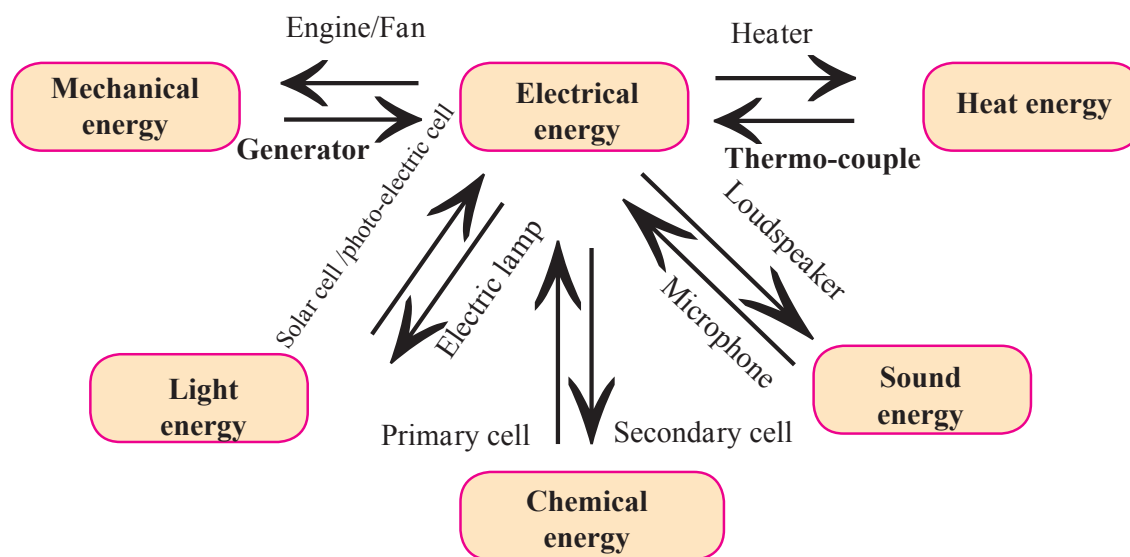


Can you tell?

Which are the different forms of energy? Which type of energy is used in each of the following processes?

1. A stretched piece of rubber 2. Fast moving car 3. The whistling of a cooker due to steam 4. The crackers exploded in Diwali 5. A fan running on electricity 6. Drawing out pieces of iron from garbage, using a magnet 7. Breaking of a glass window pane because of a loud noise.

Energy can be transformed from one type to another. For example, the exploding firecrackers convert the chemical energy stored in them into light, sound and heat energy.



2.5 Transformation of energy

Observe the above diagram (figure 2.5) and discuss how transformation of energy takes place, giving example of each.

Law of conservation of energy

‘Energy can neither be created nor destroyed. It can be converted from one form into another. Thus, the total amount of energy in the universe remains constant’.

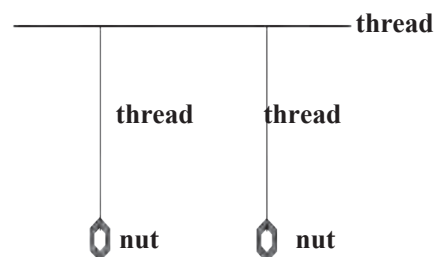


Try this

Make two pendulums of the same length with the help of thread and two nuts. Tie another thread in the horizontal position.

Tie the two pendulums to this horizontal thread in such a way that they will not hit each other while swinging. Now swing one of the pendulums and observe. What do you see?

You will see that as the speed of oscillation of the pendulum slowly decreases, the second pendulum which was initially stationary, begins to swing. Thus, one pendulum transfers its energy to the other.



2.6 Coupled oscillators

Free fall

If we release an object from a height, it gets pulled towards the earth because of the gravitational force. An object falling solely under the influence of gravitational force is said to be in free fall or to be falling freely. Let us look at the kinetic and potential energies of an object of mass m , falling freely from height h , when the object is at different heights

As shown in the figure, the point A is at a height h from the ground. Let the point B be at a distance x , vertically below A. Let the point C be on the ground directly below A and B. Let us calculate the energies of the object at A, B and C.

1. When the object is stationary at A, its initial velocity is $u = 0$

$$\therefore \text{K.E.} = \frac{1}{2} \text{mass} \times \text{velocity}^2$$

$$= \frac{1}{2} m u^2$$

$$\text{K.E.} = 0$$

$$\text{P.E.} = mgh$$

$$\therefore \text{Total energy} = \text{K.E.} + \text{P.E.}$$

$$= 0 + mgh$$

$$\text{Total Energy} = mgh \text{ --- (1)}$$

2. Let the velocity of the object be v_B when it reaches point B, having fallen through a distance x .

$$u = 0, s = x, a = g$$

$$v^2 = u^2 + 2as$$

$$v_B^2 = 0 + 2gx$$

$$v_B^2 = 2gx$$

$$\therefore \text{K.E.} = \frac{1}{2} m v_B^2 = \frac{1}{2} m (2gx)$$

$$\text{K.E.} = mgx$$

$$\text{Height of the object when at B} = h - x$$

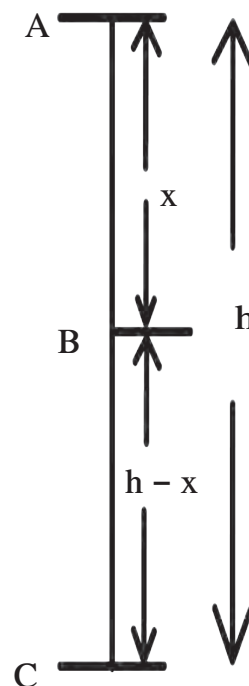
$$\therefore \text{P.E.} = mg(h - x)$$

$$\text{P.E.} = mgh - mgx$$

$$\therefore \text{Total Energy T.E.} = \text{K.E.} + \text{P.E.}$$

$$= mgx + mgh - mgx$$

$$\therefore \text{T.E.} = mgh \text{ --- (2)}$$



2.7 Free fall

3. Let the velocity of the object be v_C when it reaches the ground, near point C.

$$u = 0, s = h, a = g$$

$$v^2 = u^2 + 2as$$

$$v_C^2 = 0 + 2gh$$

$$\therefore \text{K.E.} = \frac{1}{2} m v_C^2 = \frac{1}{2} m (2gh)$$

$$\text{K.E.} = mgh$$

The height of the object from the ground at point C is

$$h = 0$$

$$\therefore \text{P.E.} = mgh = 0$$

$$\therefore \text{T.E.} = \text{K.E.} + \text{P.E.}$$

$$\text{T.E.} = mgh \text{ --- (3)}$$

From equations (1), (2) and (3) we see that the total energy of the object is the same at the three points A, B and C.

Thus, every object has potential energy when it is at a height above the ground and it keeps getting converted to kinetic energy as the object falls towards the ground. On reaching the ground (point C), all the potential energy gets converted to kinetic energy. But at any point during the fall the total energy remains constant.

$$\text{i.e., T.E.} = \text{P.E.} + \text{K.E.}$$

$$\text{T.E. at A} = mgh + 0 = mgh$$

$$\text{T.E. at B} = mgx + mg(h - x) = mgh$$

$$\text{T.E. at C} = 0 + mgh = mgh$$



Power



Think before you answer

1. Can your father climb stairs as fast as you can?
2. Will you fill the overhead water tank with the help of a bucket or an electric motor?
3. Suppose Rajashree, Yash and Ranjeet have to reach the top of a small hill. Rajashree went by car, Yash went cycling while Ranjeet went walking. If all of them choose the same path, who will reach first and who will reach last?

In the above examples, the work done is the same in each example but the time taken to perform the work is different for each person or each method. The fast or slow rate of the work done is expressed in terms of power. 'Power is the rate at which work is done.'

If W amount of work is done in time t then,

$$\text{Power} = \frac{\text{Work}}{\text{Time}} \quad P = \frac{W}{t}$$

In SI system the unit of work is J, so the unit of power is

J/s. This is called watt

$$1 \text{ watt} = 1 \text{ joule} / 1 \text{ second}$$

In the industrial sector the unit used to measure power is called 'horse power.'

$$1 \text{ horse power} = 746 \text{ watt.}$$

The unit of energy for commercial use is kilo watt hour.

1000 joule work performed in 1 second is 1 kilowatt power.

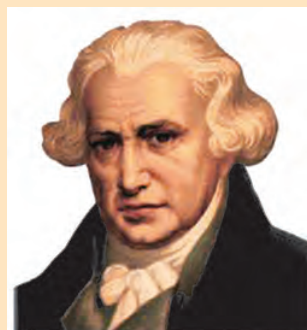
$$\begin{aligned} 1 \text{ kW h} &= 1 \text{ kW} \times 1 \text{ hr} \\ &= 1000 \text{ W} \times 3600 \text{ s} \\ &= 3600000 \text{ J} \end{aligned}$$

$$1 \text{ kW h} = 3.6 \times 10^6 \text{ J}$$

Electricity used for domestic purposes is measured in units of kilowatt hour.

$$1 \text{ kW h} = 1 \text{ unit}$$

An introduction to scientists



The steam engine was invented in 1781 by the Scottish scientist James Watt (1736 – 1819). This invention brought about an industrial revolution. The unit of power is called Watt in his honour. James Watt was the first to use the term 'horse-power'.

Solved problems

Example 1 : Swaralee takes 20 s to carry a bag weighing 20 kg to a height of 5 m. How much power has she used?

Given : $m = 20 \text{ kg}$, $h = 5 \text{ m}$, $t = 40 \text{ s}$

$\therefore$ The force which has to be applied by Swaralee,

$$F = mg = 20 \times 9.8$$

$$F = 196 \text{ N}$$

Work done by Swaralee to carry the bag to a height of 5 m,

$$W = F s = 196 \times 5 = 980 \text{ J}$$

$$\therefore \text{ power} = (P) = \frac{\text{Work}}{t} = \frac{980}{40}$$

$$P = 24.5 \text{ W}$$

Example 2 : A 25 W electric bulb is used for 10 hours every day. How much electricity does it consume each day?

Given :

$$P = 25, W = 0.025 \text{ kW}$$

$$\therefore \text{ Energy consumed} = \text{power} \times \text{time} \\ = 0.025 \times 10$$

$$\text{Energy} = 0.25 \text{ kW hr}$$

Websites for more information :

www.physicscatalyst.com

www.tryscience.org

Exercises



1. Write detailed answers?

- Explain the difference between potential energy and kinetic energy.
- Derive the formula for the kinetic energy of an object of mass m , moving with velocity v .
- Prove that the kinetic energy of a freely falling object on reaching the ground is nothing but the transformation of its initial potential energy.
- Determine the amount of work done when an object is displaced at an angle of 30° with respect to the direction of the applied force.
- If an object has 0 momentum, does it have kinetic energy? Explain your answer.
- Why is the work done on an object moving with uniform circular motion zero?

2. Choose one or more correct alternatives.

- For work to be performed, energy must be
 - transferred from one place to another
 - concentrated
 - transformed from one type to another
 - destroyed
- Joule is the unit of ...
 - force
 - work
 - power
 - energy
- Which of the forces involved in dragging a heavy object on a smooth, horizontal surface, have the same magnitude?
 - the horizontal applied force
 - gravitational force
 - reaction force in vertical direction
 - force of friction
- Power is a measure of the
 - the rapidity with which work is done
 - amount of energy required to perform the work
 - The slowness with which work is performed
 - length of time

- e. While dragging or lifting an object, negative work is done by
 (i) the applied force (ii) gravitational force (iii) frictional force (iv) reaction force

3. Rewrite the following sentences using proper alternative.

- a. The potential energy of your body is least when you are
 (i) sitting on a chair (ii) sitting on the ground (iii) sleeping on the ground (iv) standing on the ground
- b. The total energy of an object falling freely towards the ground ...
 (i) decreases (ii) remains unchanged (iii) increases (iv) increases in the beginning and then decreases
- c. If we increase the velocity of a car moving on a flat surface to four times its original speed, its potential energy
 (i) will be twice its original energy (ii) will not change (iii) will be 4 times its original energy (iv) will be 16 times its original energy.
- d. The work done on an object does not depend on
 (i) displacement (ii) applied force (iii) initial velocity of the object (iv) the angle between force and displacement.

4. Study the following activity and answer the questions.

- Take two aluminium channels of different lengths.
- Place the lower ends of the channels on the floor and hold their upper ends at the same height.
- Now take two balls of the same size and weight and release them from the top end of the channels. They will roll down and cover the same distance.

Questions

- At the moment of releasing the balls, which energy do the balls have?
- As the balls roll down which energy is converted into which other form of energy?
- Why do the balls cover the same distance on rolling down?
- What is the form of the eventual total energy of the balls?
- Which law related to energy does the above activity demonstrate ? Explain.

5. Solve the following examples.

- a. An electric pump has 2 kW power. How much water will the pump lift every minute to a height of 10 m?
(Ans : 1224.5 kg)
- b. If a 1200 W electric iron is used daily for 30 minutes, how much total electricity is consumed in the month of April? **(Ans :18 Unit)**
- c. If the energy of a ball falling from a height of 10 metres is reduced by 40%, how high will it rebound?
(Ans : 6 m)
- d. The velocity of a car increases from 54 km/hr to 72 km/hr. How much is the work done if the mass of the car is 1500 kg ?
(Ans. : 131250 J)
- e. Ravi applied a force of 10 N and moved a book 30 cm in the direction of the force. How much was the work done by Ravi?

(Ans: 3 J)

Project :

Study the various instances of transformation of energy you see around you and discuss them in class.



3. Current Electricity



- Potential and potential difference
- Conductors and insulators (bad conductors)
- Electrical resistance and Ohm's Law
- Connection of resistors and effective resistance



All around us

Electricity is of utmost importance in the modern world. We depend on it for almost everything in our day to day life. In order to avoid the inconvenience faced due to failure of power supply, hospitals, banks, offices and private institutions make alternative arrangements with the help of generators. Electricity is used for running electric furnaces, electric motors and several other instruments used in industries.

Domestic appliances like the fridge, electric oven, mixer, fans, washing machines, vacuum cleaner, rotimaker, etc. have helped us by saving time and labour. All these devices cannot be run without electricity.

Not only human beings but some animals also use electricity. For example, fishes such as eels use electricity to catch their prey and also for self-defence. The lightning that strikes the earth is an excellent example of natural flow of electricity. What if we could collect and store this electricity!



Can you recall?

You must have seen a waterfall. Which way does the water flow?

For the generation of electricity, water is released from a dam at a higher level and because of gravity, it falls to a lower level. Thus, as we know, the direction of flow of water between two points depends on the level of the two points.

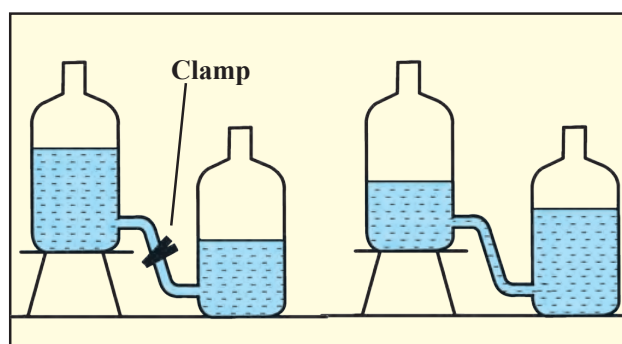
Potential and potential difference



Try this

Equipment : Two plastic bottles, rubber tube, clamp, water.

Procedure : Set up the experiment as shown in figure 3.1. Then remove the clamp from the rubber tube. Note your observations. Answer the following questions.



3.1 Level of water and direction of flow

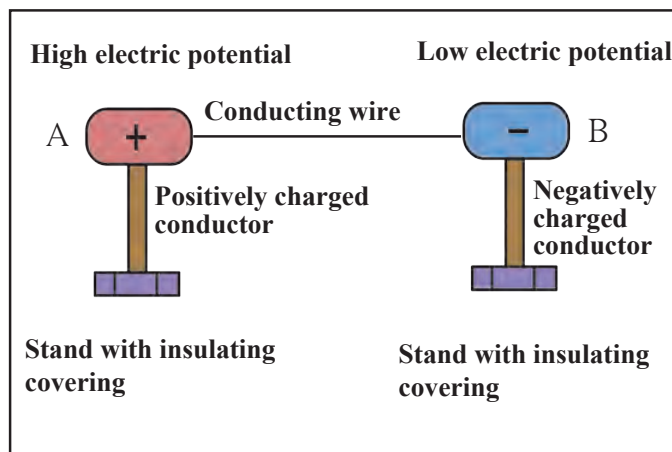
1. What happens when the clamp is removed?
2. Does the water stop flowing? Why?
3. What will you do to keep the water flowing for a longer duration?

Just like water, the flow of electric charge between two points depends on a kind of electric level at those points. This level is called electric potential.

A positive charge flows from a point of higher potential to a point of lower potential. We have seen earlier that electricity flows due to the conduction of negatively charged electrons. Electrons flow from the point of lower potential to a point of higher potential. A lightning strike is the flow of electrons from point of lower (negative) potential on the clouds to the point of higher (zero) potential on the earth. We shall study the definition of electric potential in higher standards.

The difference between the values of potentials at two points A and B is called the potential difference between them.

In the figure 3.2, conductor A is at a higher potential than conductor B. When these two are connected by a conducting wire, a potential difference is created between its two ends and electrons will flow from B to A through the wire. This flow will continue until the two conductors, A and B have the same potential, i.e. until their potential difference becomes zero. Only then will the flow of electrons stop.



3.2 Potential difference and flow of electricity

Work has to be done against the electric field to take a positive charge from a point of lower potential to a point of higher potential.

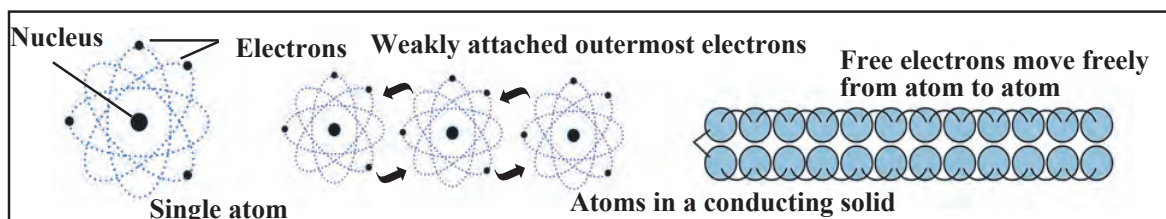
Potential difference of a cell

The difference in potential between the positive and negative terminals of a cell is the potential difference of that cell. This potential difference is caused by chemical reactions occurring inside the cell. The potential difference sets the electrons in motion and results in the flow of electricity through a conducting wire connected to the two ends of the cell.

The amount of work done to carry a unit positive charge from point A to point B is called the electric potential difference between the two points.

$$\text{Potential difference between two points} = \frac{\text{Work}}{\text{Total charge transferred}}, V = \frac{W}{Q}$$

$$1 \text{ V} = \frac{1 \text{ J}}{1 \text{ C}} \quad \text{The unit of potential difference in SI system is volt.}$$



3.3 Free electrons



An introduction to scientists

The Italian scientist Alessandro Volta constructed the first electric cell. The unit of potential difference is named 'volt' in his honour.

Volta's simple electric cell



Do you know?

Very small values of potential difference are expressed in the following units.

1. 1mV (millivolt) = 10^{-3} V
2. 1 μ V (microvolt) = 10^{-6} V

Large values of potential difference are expressed in the following units.

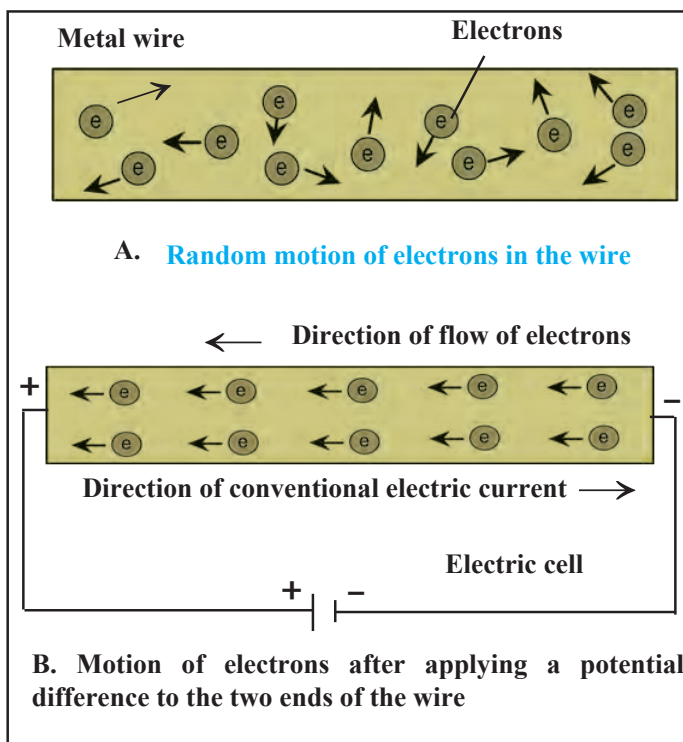
1. 1kV (kilovolt) = 10^3 V
2. 1MV (megavolt) = 10^6 V

Free electrons : Every atom of a metallic conductor has one or more outermost electrons which are very weakly bound to the nucleus. These are called free electrons.

As shown in figure 3.3, these electrons can easily move from one part of a conductor to its other parts. The negative charge of the electrons also gets transferred as a result of this motion. The free electrons in a conductor are the carriers of negative charge.

Current flowing through a wire

As shown in the figure 3.4 A, if a conducting wire is not connected to a cell, its free electrons move randomly in all directions in the space between the atoms. When we connect the ends of the wire to the two terminals of a cell, electric force acts on the electrons. Being negatively charged, they start moving from the negative (lower potential) to the positive (higher potential) terminal of the cell, as shown in figure 3.4 B. Due to the flow of these electrons, current starts to flow through the wire. This motion of electrons is irregular but there is a definite, non-zero value to their average velocity.



3.4 Motion of free electrons

Electric current

An electric current is the flow of electrons through a conductor. Quantitatively, current (I) is defined as the charge passing through a conductor in unit time.

If charge Q is flowing through cross-section of a conductor in time t then the

$$\text{current} = I = \frac{Q}{t}$$

The unit of charge in SI units is Coulomb (C). Current is expressed in Ampere (A). The charge of one electron is 1.6×10^{-19} C.

Ampere : One ampere current is said to flow in a conductor if one Coulomb charge flows through it every second.

$$1\text{A} = \frac{1\text{C}}{1\text{s}}$$



Do you know?

Very small values of current are expressed in the following units.

1. 1mA (mili ampere) = 10^{-3} A
2. 1 μ A (micro ampere) = 10^{-6} A

The French mathematician and scientist, Ampere, conducted a number of experiments on electricity. Today, we can measure the current flowing in a conductor only because of his work. The unit of current is called ampere in recognition of his work.



Example : A current of 0.4 A flows through a conductor for 5 minutes. How much charge would have passed through the conductor?

Given : I = 0.4 A

$$t = 5 \text{ min} = 5 \times 60 \text{ s} = 300 \text{ s}$$

Formula $Q = I \times t$

$$Q = 0.4 \text{ A} \times 300 \text{ s}$$

$$Q = 120 \text{ C.}$$

$\therefore$ Charge passing through the conductor = 120 C

Even though, electrons move from negative end to positive end, conventionally, the direction of current flow is taken to be opposite of the direction of flow of electrons i.e. from the +ve end to the -ve end of a cell.

Using ICT

Study current electricity and other scientific concepts with the help of simulation technology .

Website :

www.phet.colorado.edu

www.edumedia-sciences.com

Find some more websites like the ones mentioned above and share them with others.

Resistance and Ohm's Law

Ohm's law

The relationship between the current flowing through a wire (I) and the potential difference across its ends (V) can be obtained from the law that was given by the German scientist George Simon Ohm.

If the physical state of a conductor remains constant, the current (I) flowing through it is directly proportional to the potential difference (V) between its two ends.

$$I \propto V$$

$$I = kV \quad (k = \text{constant of proportionality})$$

$$I \times \frac{1}{k} = V \quad \left(\frac{1}{k} = R = \text{Resistance of the conductor} \right)$$

$$I \times R = V \quad \text{Hence } V = IR \text{ or } R = \frac{V}{I}$$

This is known as Ohm's law.

Physical state of a conductor means its length, area of cross-section, temperature and the material it is made up of.

We can obtain the SI unit of resistance from the above formula. Potential difference and current are measured in Volts and Amperes respectively. The unit of resistance is called Ohm. It is indicated by the symbol Ω .

$$\therefore \frac{1 \text{ Volt}}{1 \text{ Ampere}} = 1 \text{ Ohm } (\Omega)$$

Resistance of one Ohm : If one Ampere current flows through a conductor when one Volt potential difference is applied between its ends, then the resistance of the conductor is one Ohm.

Resistance and resistivity of a conductor

As shown in figure 3.4, there are a large number of free electrons in a conductor. They are constantly in random motion. When a potential difference is applied between the two ends of the conductor, these electrons start moving from the end at lower potential to the end at higher potential. This directional motion of the electrons causes the flow of current. Moving electrons strike the atoms and ions which lie along their path. Such collisions cause hindrance to the flow of electrons and oppose the current. This hindrance is called the resistance of the conductor.

Resistivity : At a given temperature, the resistance (R) of a conductor depends on its length (L), area of cross-section (A) and the material it is made of.



German physicist, George Simon Ohm established a law for measuring the resistance of a conductor. In his honour, the unit of resistance is called the Ohm.

If the resistance of a conductor is R , then

$$R \propto L$$

Also, $R \propto \frac{1}{A}$

$$\therefore R \propto \frac{L}{A}$$

$$R = \rho \frac{L}{A}$$

Think about it

How will you prove that the unit of resistivity is $\Omega \text{ m}$?

Resistivity of some materials

Copper - $1.7 \times 10^{-8} \Omega \text{ m}$

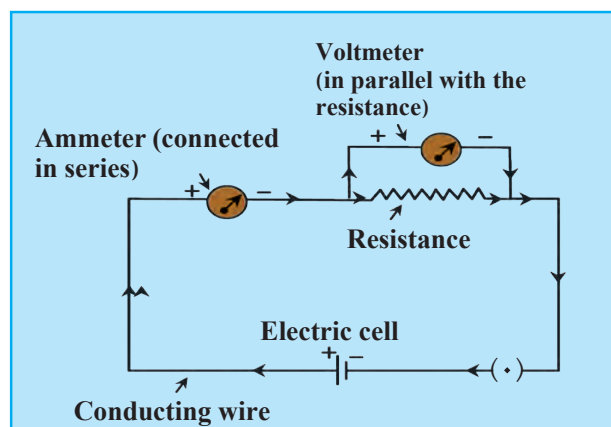
Nichrome - $1.1 \times 10^{-6} \Omega \text{ m}$

Diamond - 1.62×10^{13} to $1.62 \times 10^{18} \Omega \text{ m}$

ρ is the constant of proportionality and is called the resistivity of the material. The unit of resistivity in SI units is Ohm metre ($\Omega \text{ m}$). Resistivity is a specific property of a material and different materials have different resistivity

Electric circuit

A continuous path of an electric current through conducting wires connected to the two ends of a cell and other resistances is called an electric circuit. A circuit is depicted by a figure. This figure shows how different components are to be connected in the circuit, by using special symbols for each of the components. Such a figure is called an electric circuit diagram.

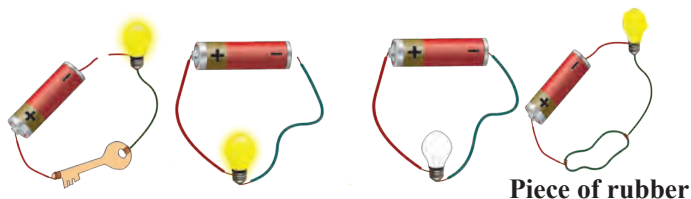


3.5 Electrical circuit

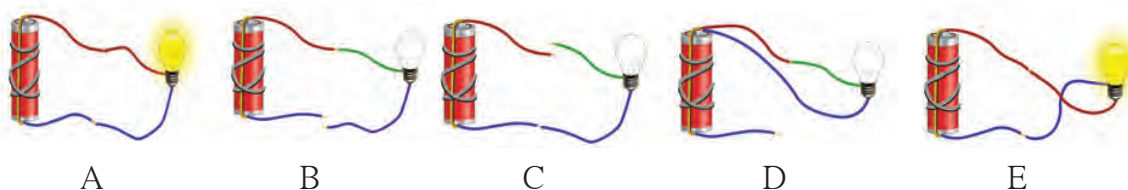
In the circuit in figure 3.5, an ammeter is used to measure current and a voltmeter to measure the potential difference between the two ends of a resistor. As the voltmeter has a very high resistance, only very small current flows through it.



Observe



1. Point out the mistakes in the figure above.
2. Why are the bulbs in Figures B, C and D below not lighting up ?



Symbols for components of an electric circuit and their uses

Component	Picture	Symbol	Use
Electric cell			To apply a potential difference between two ends of a conductor
Battery (collection of a number of cells)			To apply a larger potential difference between two ends of a conductor
Open tap key or plug key			To stop the flow of current flowing in a circuit by disconnecting two ends of a wire
Closed tap key or plug key			To start the flow of current in a circuit by connecting two ends of a wires.
Connecting (conducting) wires			To connect various components in the circuit.
Crossing wires			To show wires which cross but are not connected
Light bulb			To test the flow of electricity : Lighted : current is flowing; unlighted : current is not flowing
Resistance			To control the flow of current in the circuit
Variable resistance			To change the resistance as required and thereby control the current
Ammeter			To measure the current flowing in the circuit
Voltmeter			To measure the potential difference between two points in the circuit

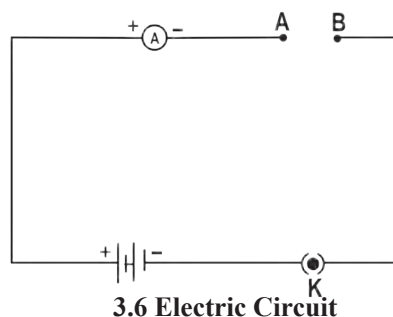


Try this

Material: copper and aluminium wires, glass rod, rubber

Method : Make connections as shown in figure 3.6.

First connect a copper wire between points A and B and measure the current in the circuit. Then, in place of the copper wire, connect aluminium wire, glass rod, rubber, etc, one at a time and measure the current each time. Compare the values of the current in different cases.



3.6 Electric Circuit

Conductors and insulators

We have learnt about the resistance to an electric current. We can divide substances into conductors and insulators (bad conductors).

Conductors : Those substances which have very low resistance are called conductors. Current can flow easily through such materials.

Insulators : Those substances which have extremely high resistance and through which current cannot flow are called insulators.

1. Why are some substances conductors while others are insulators?
2. Why can our body conduct electricity?

Make a list of conductors and insulators you see around you.

Experimental proof of Ohm's law :

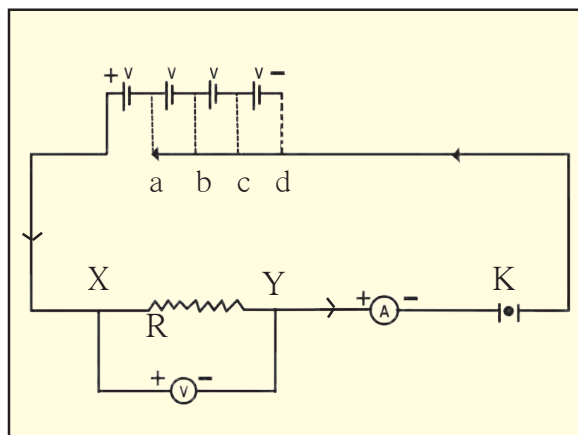


Try this

Material : 4 cells of 1.5 V each, ammeter, voltmeter, conducting wires, nichrome wire, plug key.

Procedure :

1. Set up the circuit as shown in figure 3.7.
2. Use the nichrome wire XY as the resistance.
3. Connect one of the 4 cells as shown in figure 3.7 (a.) Take readings of ammeter and voltmeter and enter them in the table below.
4. Now add the rest of the cells one by one as shown in figures 3.7 (b, c and d). Enter the readings in the table for each case.
5. Determine the values of $\frac{V}{I}$ for each case.
6. Draw a graph between current and potential difference and study it.



3.7 Verification of Ohm's Law

Observation chart

No.	Number of cells used	Current (I) (mA)	Current (I) (A)	Potential difference (Volts)	$V / I = R$ in (Ω)
1.					
2.					
3.					
4.					

Solved examples : Ohm's law and resistivity

Example 1 : The resistance of the filament in a light bulb is 1000Ω . If the bulb is fed by a current from a source of potential difference 230 V , how much current will flow through it?

Given : $R = 1000 \Omega$
 $V = 230 \text{ V}$

Formula $I = \frac{V}{R}$
 $\therefore I = \frac{230 \text{ V}}{1000 \Omega} = 0.23 \text{ A}.$

$\therefore$ The current flowing through the filament of the bulb $= 0.23 \text{ A}$.

Example 2 : The length of a conducting wire is 50 cm and its radius is 0.5 mm . If its resistance is 30Ω , what is the resistivity of its material?

Given : $L = 50 \text{ cm} = 50 \times 10^{-2} \text{ m}$
 $r = 0.5 \text{ mm} = 0.5 \times 10^{-3} \text{ m}$
 $= 5 \times 10^{-4} \text{ m}$ and $R = 30 \Omega$
 Resistivity $\rho = \frac{RA}{L}$
 and $A = \pi r^2$

$$\begin{aligned} \therefore \rho &= R \frac{\pi r^2}{L} \\ &= \frac{30 \times 3.14 \times (5 \times 10^{-4})^2}{50 \times 10^{-2}} \\ &= \frac{30 \times 3.14 \times 25 \times 10^{-8}}{50 \times 10^{-2}} \\ &= 47.1 \times 10^{-6} \Omega \text{ m} \\ &= 4.71 \times 10^{-5} \Omega \text{ m} \\ \therefore \text{Resistivity of the wire } &4.71 \times 10^{-5} \Omega \text{ m} \end{aligned}$$

Example 3 : A current of 0.24 A flows through a conductor when a potential difference of 24 V is applied between its two ends. What is its resistance?

Given : $V = 24 \text{ V}$, $I = 0.24 \text{ A}$

Formula $R = \frac{V}{I}$
 $\therefore I = \frac{24 \text{ V}}{0.24 \text{ A}}$
 $R = 100 \Omega$

The resistance of the conductor is 100Ω .

Example 4 : Determine the current that will flow when a potential difference of 33 V is applied between two ends of an appliance having a resistance of 110Ω . If the same current is to flow through an appliance having a resistance of 500Ω , how much potential difference should be applied across its two ends?

Given : $V = 33 \text{ V}$ and $R = 110 \Omega$

In the first case,

$$I = \frac{V}{R} = \frac{33}{110}$$

$$\therefore I = 0.3 \text{ A}$$

Current flowing through the appliance $= 0.3 \text{ A}$

In the second case,

$$I = 0.3 \text{ A}, R = 500 \Omega$$

$$V = IR = 0.3 \times 500 \text{ V} = 150 \text{ V}.$$

The required potential difference between its two ends $= 150 \text{ V}$

Using ICT

Using the internet find out about the different softwares used to solve mathematical problems and use them to solve problems given in this and other chapters.

Example 5 : Determine the resistance of a copper wire having a length of 1 km and diameter of 0.5 mm

Given : ρ = Resistivity of copper = $1.7 \times 10^{-8} \Omega \text{ m}$

Converting all measures into metres.

$$L = 1 \text{ km} = 1000 \text{ m} = 10^3 \text{ m}$$

$$d = 0.5 \text{ mm} = 0.5 \times 10^{-3} \text{ m}$$

If r is the radius of the wire then, its area of cross-section

$$\begin{aligned} A &= \pi r^2 \\ \therefore A &= \pi \times \left(\frac{d}{2}\right)^2 \\ &= \frac{\pi}{4} (0.5 \times 10^{-3})^2 \text{ m}^2 = 0.2 \times 10^{-6} \text{ m}^2 \\ R &= \rho \frac{L}{A} = \frac{1.7 \times 10^{-8} \Omega \text{ m} \times (10^3 \text{ m})}{0.2 \times 10^{-6} \text{ m}^2} = 85 \Omega \end{aligned}$$

System of resistors and effective resistance

A resistor is a two ended component having a given amount of resistance between its two ends. In several electrical devices, a number of resistors are connected together in different ways. Ohm's law is applicable to all such connected resistors.

Resistors in series

Study figure 3.8.

You can see that the ends of the three resistors are connected so that they follow one after the other in a single line. These resistors are said to be connected in 'series.' In such an arrangement, the same current flows through each resistor. The value of current is I and the potential difference between C and D is V .

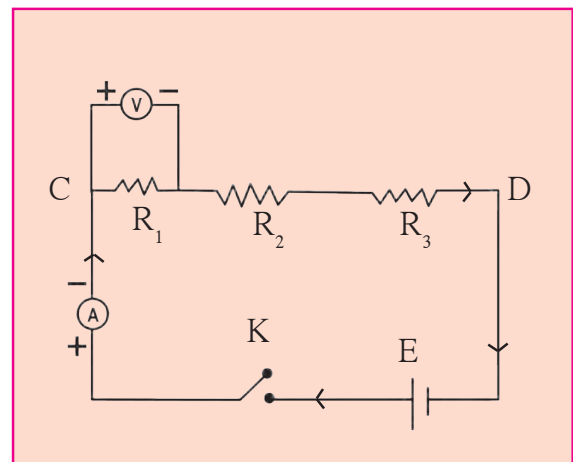
The three resistors, R_1 , R_2 and R_3 are connected in series in the circuit. If V_1 , V_2 , V_3 are potential differences of every resistor R_1 , R_2 and R_3 respectively, then

$$V = V_1 + V_2 + V_3 \text{ -----(1)}$$

If R_s (S for series) is the effective resistance between C and D, then, according to Ohm's law,

$$V = I R_s$$

$V_1 = I R_1$, $V_2 = I R_2$ and $V_3 = I R_3$. Substituting all these in equation (1) we get,



3.8 Resistors in series

$$I R_s = I R_1 + I R_2 + I R_3$$

$$R_s = R_1 + R_2 + R_3$$

If n resistors are connected in series then,

$$R_s = R_1 + R_2 + R_3 + \text{-----} + R_n$$

If resistors are connected in series,

1. The same current flows through each resistor.
2. The effective resistance of the resistors is equal to the sum of their individual resistances.
3. The potential difference between the two extremes of the arrangement is equal to the sum of the potential differences across individual resistors.
4. The effective resistance is larger than each of the individual resistances.
5. This arrangement is used to increase the resistance in a circuit.

**Do you know?**

When resistors are connected in series, they are connected one after another. If any one of them does not function, the circuit breaks and there is no flow of electricity. If two bulbs are connected in series, they will both give less light than if they had been connected individually. If three bulbs are connected in series, their light output will decrease further.

Think about it : What is the reason behind this?

Solved examples for resistors in series

Example 1 : Three resistors having resistances of $15\ \Omega$, $3\ \Omega$ and $4\ \Omega$ are connected in series. What is the effective resistance in the circuit?

Given : $R_1 = 15\ \Omega$, $R_2 = 3\ \Omega$, $R_3 = 4\ \Omega$

Effective resistance $R_s = R_1 + R_2 + R_3 = 15 + 3 + 4 = 22\ \Omega$

$\therefore$ The effective resistance in the circuit = $22\ \Omega$

Example 2 : Two resistors having resistances of $16\ \Omega$ and $14\ \Omega$ are connected in series. If a potential difference of $18\ \text{V}$ is applied across them, calculate the current flowing through the circuit and the potential difference across the end of each individual resistor.

Given : $R_1 = 16\ \Omega$ and $R_2 = 14\ \Omega$

$R_s = 14\ \Omega + 16\ \Omega = 30\ \Omega$

Let the current flowing through the circuit be I and the potential differences across the end $16\ \Omega$ and $14\ \Omega$ be V_1 and V_2 respectively.

$$V = IR \quad V = V_1 + V_2 = 18\ \text{V}$$

$$I = \frac{V}{R} = \frac{18\ \text{V}}{30\ \Omega}$$

$$\therefore I = 0.6\ \text{A.}$$

$$V_1 = IR_1$$

$$V_1 = 0.6 \times 16 = 9.6\ \text{V}$$

$$V_2 = IR_2 = 0.6 \times 14 = 8.4\ \text{V}$$

$\therefore$ The current flowing in the circuit is $0.6\ \text{A}$ and the potential differences across the end of the resistances $16\ \Omega$ and $14\ \Omega$ are $9.6\ \text{V}$ and $8.4\ \text{V}$ respectively.



Do you know?

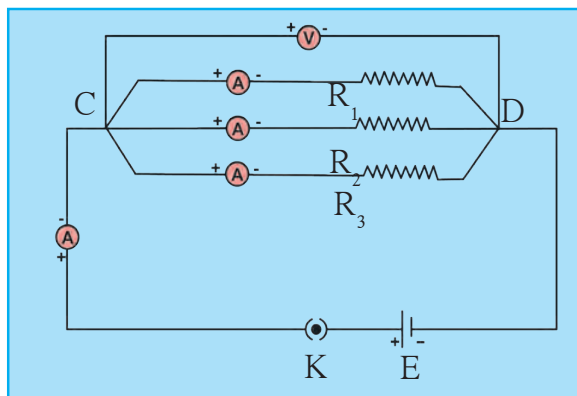
The resistance of some conductors becomes nearly zero if their temperature is decreased upto a certain value close to 0 Kelvin (K). Such conductors are called super conductors.

Some conductors do not obey Ohm's law. Such conductors are called non-ohmic conductors.

Resistors in parallel

Resistors are said to be connected in parallel when their ends are connected at both sides as shown in figure 3.9. The figure shows three resistors R_1 , R_2 and R_3 connected in parallel between points C and D. V is the potential difference between C and D. Let I_1 , I_2 and I_3 be the currents flowing through R_1 , R_2 , and R_3 respectively.

Then, the total current flowing through the circuit is



3.9 Resistors in parallel

$$I = I_1 + I_2 + I_3 \text{-----(1)}$$

Let R_p (p for parallel) be the effective resistance between C and D. According to Ohm's law,

$$I = \frac{V}{R_p} \quad \text{Similarly } I_1 = \frac{V}{R_1}, \quad I_2 = \frac{V}{R_2}, \quad I_3 = \frac{V}{R_3}$$

Substituting these in equation (1)

$$\begin{aligned} \frac{V}{R_p} &= \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} \\ \therefore \frac{1}{R_p} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \end{aligned}$$

If n resistors are connected in parallel,

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$

Even if any one of the several bulbs connected in parallel becomes non-functional because of some damage to its filament, the circuit does not break as the current flows through the other paths, and the rest of the bulbs light up.

When several bulbs are connected in parallel, they emit the same amount of light as when they are connected individually in the circuit, while bulbs connected in series emit less light than when connected individually.

If a number of resistors are connected in parallel,

1. The inverse of the effective resistance is equal to the sum of the inverses of individual resistances.
2. The current flowing through an individual resistor is proportional to the inverse of its resistance and the total current flowing through the circuit is the sum of the currents flowing through individual resistors.
3. The potential difference across the end of all resistors is the same.
4. The effective resistance of resistors connected in parallel is less than the least resistance of individual resistors.
5. This arrangement is used to reduce the resistance in a circuit.

Solved examples of resistors in parallel

Example 1: Resistors having resistances of $15\ \Omega$, $20\ \Omega$ and $10\ \Omega$ are connected in parallel. What is the effective resistance in the circuit?

Given : $R_1 = 15\ \Omega$, $R_2 = 20\ \Omega$ and $R_3 = 10\ \Omega$

$$\begin{aligned}\frac{1}{R_p} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \\ \frac{1}{R_p} &= \frac{1}{15} + \frac{1}{20} + \frac{1}{10} = \frac{4 + 3 + 6}{60} = \frac{13}{60} \\ R_p &= \frac{60}{13} = 4.615\ \Omega\end{aligned}$$

$\therefore$ Effective resistance in the circuit = $4.615\ \Omega$, less than the least of the three, $10\ \Omega$.

Example 2 : Three resistors having resistances of $5\ \Omega$, $10\ \Omega$ and $30\ \Omega$ are connected in parallel and a potential difference of $12\ \text{V}$ is applied across them. Obtain the current flowing through the circuit and through individual resistors. What is the effective resistance in the circuit?

Given : $R_1 = 5\ \Omega$, $R_2 = 10\ \Omega$ and $R_3 = 30\ \Omega$, $V = 12\ \text{V}$

$$\begin{aligned}I_1 &= \frac{V}{R_1} = \frac{12}{5} = 2.4\ \text{A} \\ I_2 &= \frac{V}{R_2} = \frac{12}{10} = 1.2\ \text{A} \\ I_3 &= \frac{V}{R_3} = \frac{12}{30} = 0.4\ \text{A}\end{aligned}$$

$$I = I_1 + I_2 + I_3 = 2.4 + 1.2 + 0.4 = 4.0\ \text{A}$$

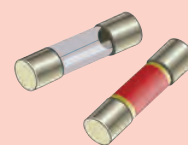
$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{5} + \frac{1}{10} + \frac{1}{30} = \frac{6 + 3 + 1}{30} = \frac{10}{30} = \frac{1}{3}$$

Effective resistance in the circuit = $3\ \Omega$ and the current flowing through the resistances $5\ \Omega$, $10\ \Omega$ and $30\ \Omega$ is $2.4\ \text{A}$, $1.2\ \text{A}$ and $0.4\ \text{A}$ respectively. The total current flowing through the circuit is $4\ \text{A}$.

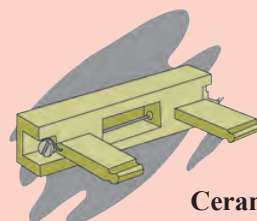
Domestic electrical connections

The electricity in our homes is brought through the main conducting cable either from the electric pole or from underground cables. Usually, there are three wires in the cable. One is called the live wire which brings in the current. It has a red or brown insulation. The other wire is called neutral wire through which the current returns. It is blue or black. In India, the voltage difference between the live and neutral wires is about 220V. Both these wires are connected to the electric meter through a fuse. Through a main switch, they are connected to all the conducting wires inside the home so as to provide electricity to every room. In each separate circuit, various electrical appliances are connected between the live and neutral wires. The different appliances are connected in parallel and the potential difference across every appliance is the same. The third wire is called the earth wire and is of yellow or green colour. This is connected to a metal plate buried deep underground near the house and is for safety purposes.

Fuse wire : Fuse wire is used to protect domestic appliances. It is made of a mixture of substances and has a specific melting point. It is connected in series to the electric appliances. If for some reason, the current in the circuit increases excessively, the fuse wire gets heated up and melts. The circuit gets broken and the flow of current stops, thus protecting the appliance. This wire is fitted in a groove in a body of porcelain - like non-conducting material. For domestic use, fuse wires with upper limits of 1 A, 2 A, 3 A, 4 A, 5 A and 10 A are used.



Cartridge fuse



Ceramic fuse

Precautions to be taken while using electricity

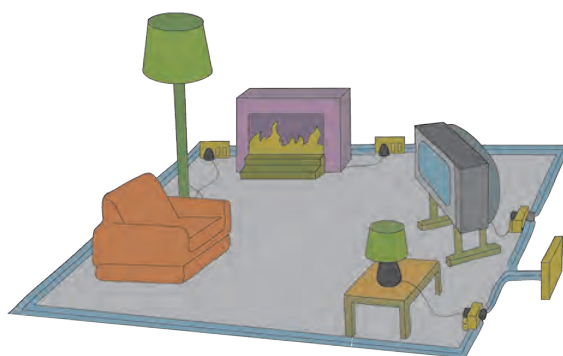
1. Electric switches and sockets should be fitted at a height at which small children cannot reach and put pins or nails inside. Plug wires should not be pulled while removing a plug from its socket. Pull a plug.
2. Before cleaning an electrical appliance it should be switched off and its plug removed from the socket.
3. One's hands should be dry while handling an electrical appliance, and, as far as possible, one should use footwear with rubber soles. As rubber is an insulator, it prevents the current from flowing through our body, thereby protecting it.
4. If a person gets an electric shock, you should not touch that person. You should switch off the main switch and if the switch is too far or you do not know where it is located, then you should remove the plug from the socket if possible. If not, then you should use a wooden pole to push the person away from the electric wire.

Exercises

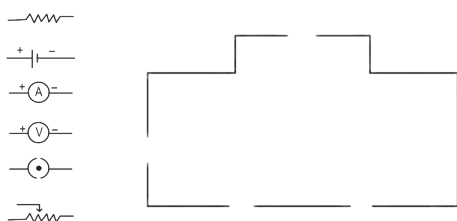


1. The accompanying figure shows some electrical appliances connected in a circuit in a house. Answer the following questions.

- By which method are the appliances connected?
- What must be the potential difference across individual appliances?
- Will the current passing through each appliance be the same? Justify your answer.
- Why are the domestic appliances connected in this way?
- If the T.V. stops working, will the other appliances also stop working? Explain your answer.



2. The following figure shows the symbols for components used in the accompanying electrical circuit. Place them at proper places and complete the circuit.



Which law can you prove with the help of the above circuit?

3. Umesh has two bulbs having resistances of $15\ \Omega$ and $30\ \Omega$. He wants to connect them in a circuit, but if he connects them one at a time the filament gets burnt. Answer the following.

- Which method should he use to connect the bulbs?
- What are the characteristics of this way of connecting the bulbs depending on the answer of question A above?

- What will be the effective resistance in the above circuit?

4. The following table shows current in Amperes and potential difference in Volts.

- Find the average resistance.
- What will be the nature of the graph between the current and potential difference? (Do not draw a graph.)
- Which law will the graph prove? Explain the law.

V (Volts)	I (Amp)
4	9
5	11.25
6	13.5

5. Match the pairs

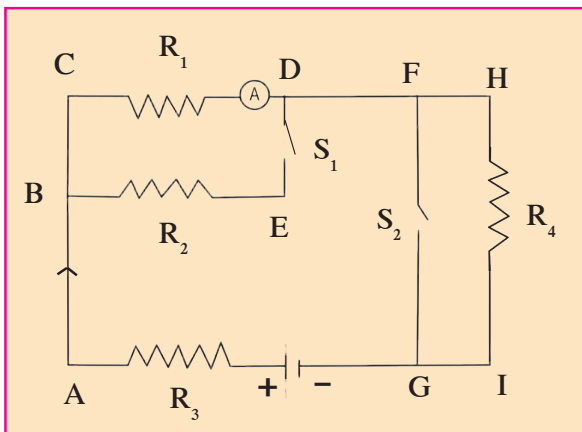
'A' Group

- Free electrons
- Current
- Resistivity
- Resistances in series

'B' Group

- V/R
- Increases the resistance in the circuit
- Weakly attached
- VA/LI

6. The resistance of a conductor of length x is r . If its area of cross-section is a , what is its resistivity? What is its unit?
7. Resistances R_1 , R_2 , R_3 and R_4 are connected as shown in the figure. S_1 and S_2 are two keys. Discuss the current flowing in the circuit in the following cases.



- Both S_1 and S_2 are closed.
 - Both S_1 and S_2 are open.
 - S_1 is closed but S_2 is open.
8. Three resistances x_1 , x_2 and x_3 are connected in a circuit in different ways. x is the effective resistance. The properties observed for these different ways of connecting x_1 , x_2 and x_3 are given below. Write the way in which they are connected in each case. (I-current, V-potential difference, x-effective resistance)

- Current I flows through x_1 , x_2 and x_3
- x is larger than x_1 , x_2 and x_3
- x is smaller than x_1 , x_2 and x_3
- The potential difference across x_1 , x_2 and x_3 is the same
- $x = x_1 + x_2 + x_3$

f. $x = \frac{1}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3}}$

9. Solve the following problems.

- A. The resistance of a 1m long nichrome wire is 6Ω . If we reduce the length of the wire to 70 cm. what will its resistance be?

(Answer : 4.2Ω)

- B. When two resistors are connected in series, their effective resistance is 80Ω . When they are connected in parallel, their effective resistance is 20Ω . What are the values of the two resistances?

(Answer : 40Ω , 40Ω)

- C. If a charge of 420 C flows through a conducting wire in 5 minutes what is the value of the current?

(Answer : 1.4 A)

Project :

Ask an electrician to explain to you the electrical wiring in your house and other related important points. Take care to understand them well, and explain them to others also.



4. Measurement of Matter



- Laws of chemical combination ➤ Atom - shape, mass, valency
- Molecular mass and the concept of mole ➤ Radicals



Can you recall?

1. What is the Dalton's atomic theory ?
2. How are the compounds formed ?
3. What are the molecular formulae of common salt, slaked lime, water, lime, limestone ?

In the previous standard we have learnt that compounds are formed by chemical combination of elements. We have also learnt that an important principle of Dalton's atomic theory is that molecules of a compound are formed by joining atoms of different elements to each other.

Laws of chemical combination

The composition of a substance changes during a chemical change. The fundamental experiments in this regard were performed by scientists in the 18th and 19th century. While doing this, they measured accurately, the substances used and formed and discovered the laws of chemical combination. Scientists could then write the molecular formulae of various compounds on the basis of Dalton's atomic theory and the laws of chemical combination. Here we shall verify the laws of chemical combination by means of known molecular formulae.



Try this

Apparatus : Conical flask, test tubes, balance, etc.

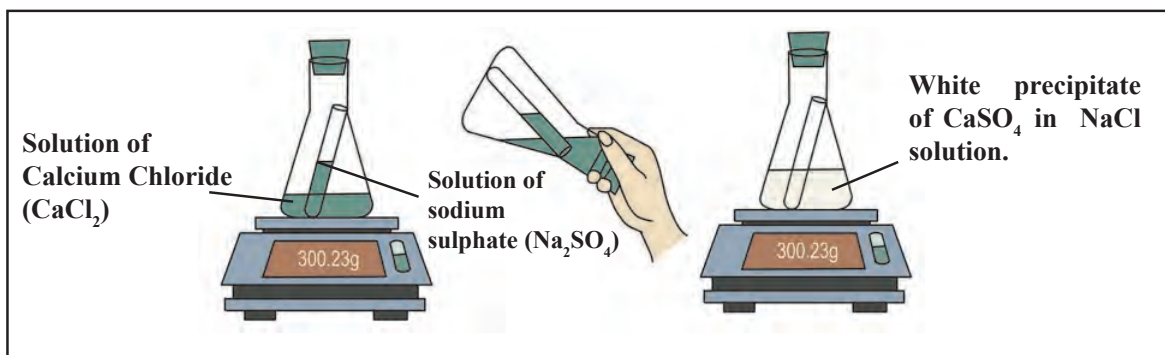
Chemicals : Calcium chloride (CaCl_2), sodium sulphate (Na_2SO_4), calcium oxide (CaO), Water (H_2O)

Activity 1

- Take 56 g calcium oxide in a large conical flask and put 18 g water in it.
- Observe what happens.
- Measure the mass of the substance formed.
- What similarity do you find? Write your inference.

Activity 2

- Take a solution of calcium chloride in a conical flask and a solution of sodium sulphate in a test tube.
- Tie a thread to the test tube and insert it in the conical flask.
- Seal the conical flask with an airtight rubber cork.
- Weigh mass of the conical flask using a balance.
- Now tilt the conical flask so that the solution in the test tube gets poured in the conical flask.
- Now weigh mass of the conical flask again.
- Which changes did you notice? Was there any change in the mass?



4.1 Verification of law of the chemical combination

Law of conservation of matter

In the above activities, the mass of the original matter and the mass of the matter newly formed as a result of the chemical change are equal. In 1785, the French Scientist Antoine Lavoisier inferred from his research that ‘there is no rise or drop in the mass of the matter during a chemical reaction.’ In a chemical reaction the total mass of the **reactants** is same as the total mass of the **products** formed due to the chemical reactions and this is called the law of conservation of matter.

Law of constant proportion

In 1794 the French scientist J. L. Proust stated the law of constant proportions as “The proportion by mass of the constituent elements in the different samples of a compound is fixed,” e.g., the proportion by mass of hydrogen and oxygen in water is obtained from any source 1:8. This means that 9 g water is formed by chemical combination of 1 g hydrogen and 8 g oxygen. Similarly, the proportion by mass of carbon and oxygen in carbon dioxide obtained from any source is 3:8. This means that in 44 g of carbondioxide there is 12 g of carbon and 32 g of oxygen so that the proportion by mass of carbon and oxygen is 3:8.



An introduction to scientists

Antoine Lavoisier (1743-94)

French scientist, father of modern chemistry, substantial contribution in the fields of chemistry, biology and economics.

- Nomenclature of oxygen and hydrogen.
- Showed that matter combines with oxygen during combustion.
- Was the first to use accurate weighing techniques to weigh mass of reactants and products in a chemical reaction.
- Discovered that water is made up of hydrogen and oxygen.
- First writer of the law of conservation of mass, in a chemical reaction.
- Assigned systematic names to the compounds, e.g. sulphuric acid, copper sulphate, sulphite.
- Authored the first book on modern chemistry- ‘Elementary Treatise on Chemistry in 1789.’
- Studied the elements such as oxygen, hydrogen, nitrogen, phosphorus, mercury, zinc and sulphur.

Verification of the law of constant proportion

Many compounds can be made by different methods. For example, two samples of the compound copper oxide, CuO , were obtained, one by decomposition of copper carbonate, CuCO_3 , and another by decomposition of copper nitrate, $\text{Cu}(\text{NO}_3)_2$. From each of these samples, a mass of 8g of copper oxide was taken and each was treated independently with hydrogen gas. Both gave 6.4 g copper and 1.8 g water. Let us see how does this is a verification of the law of constant proportion.

The reaction of copper oxide with hydrogen yielded two known substances, namely, the compound water and the element copper. It is known that, in the compound water, H_2O , the elements H and O are in the proportion 1:8 by mass. This means that in 9g water there are 8g of the element oxygen. Therefore, 1.8g water contains $(8 \times 1.8/9 = 1.6)\text{g}$ oxygen. This oxygen has come from 8g copper oxide. It means that 8g of both the samples of copper oxide contained 6.4g copper and 1.6g oxygen; and the proportion by mass of copper and oxygen in it is 6.4:1.6, that is, 4:1. Thus, the experiment showed that the proportion by mass of the constituent elements in different samples of a compound is constant.

Now let us see what the expected proportion by mass of the constituent elements of copper oxide would be from its known molecular formula CuO . To find out this, we need to use the known atomic masses of the elements. The atomic masses of Cu and O are 63.5 and 16 respectively. This means that the proportion by mass of the constituent elements Cu and O in the compound CuO is 63.5 : 16 which is 3.968:1, or approximately 4:1.

The experimental value of proportion by mass of the constituent elements matched with the expected proportion calculated from the molecular formula. Thus, the law of constant proportion is verified.

Atom : size, mass and valency



Can you recall?

1. From which experiments was it discovered that atoms have an internal structure? When?
2. What are the two parts of an atom? What are they made up of?

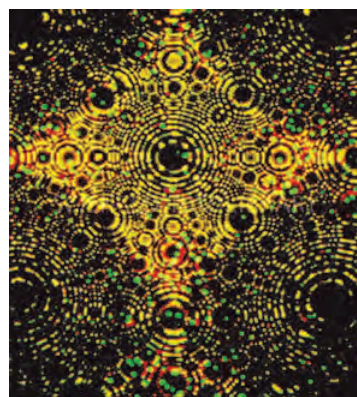
We have learnt that at the centre of an atom is the nucleus and that there are moving electrons in the extra-nuclear part. The electrons are negatively charged elementary particles while the elementary particles that make up the nucleus are positively charged protons and electrically neutral neutrons.

The size of an atom is determined by its radius. The atomic radius of an isolated atom is the distance between the nucleus of an atom and its outermost orbit. Atomic radius is expressed in nanometres.

Approximate size of atom and Molecule

Atomic radius (in metres)	Example
10^{-10}	Hydrogen atom
10^{-9}	Water molecule
10^{-8}	Haemoglobin molecule

$$\frac{1}{10^9} \text{ m} = 1 \text{ nm}$$
$$1 \text{ m} = 10^9 \text{ nm.}$$



4.2 Field ion microscope image of iridium atoms (every spot in this image is an atom)

Atoms are very tiny. Modern instruments like electron microscope, field ion microscope, scanning tunneling microscope enlarged images of the atom. Look at the image of an atom fig. 4.2 obtained with a field ion microscope.

The atomic size depends on the number of electron orbits in the atom. The greater the number of orbits the larger the size. For example, an atom of K is bigger than an atom of Na. If two atoms have the same outermost orbit, then the atom having the larger number of electrons in the outermost orbit is smaller than the one having fewer electrons in the same outermost orbit. For example an atom of Mg is smaller than an atom of Na.

The mass of an atom

The mass of an atom is concentrated in its nucleus and it is due to the protons (p) and neutrons (n) in it. The total number (of protons and neutrons) in the atomic nucleus is called the **atomic mass number**. Protons and neutrons are together called **nucleons**.

An atom is very tiny. Then how do we determine its mass? Scientists too, faced this question. It was not possible for scientists of the 19th century to measure atomic mass accurately. Therefore, the concept of 'relative mass of an atom' was put forth. A reference atom was required for expressing the relative mass of an atom. The hydrogen atom being the lightest was initially chosen as the reference atom. The relative mass of a hydrogen atom which has only one proton in its nucleus was accepted as one (1). Therefore, the magnitude of the relative atomic masses of various atoms became equal to their atomic mass number (p+n).

Let us see how to express the relative mass of a nitrogen atom, having accepted the relative atomic mass of hydrogen as 1.

The mass of one nitrogen atom is fourteen (14) times that of a hydrogen atom. Therefore, the relative mass of a nitrogen atom is 14. This is how the relative atomic masses of various elements were determined. On this scale, the relative atomic masses of many elements came out to be fractional. Therefore, in the course of time, some other atoms were chosen as reference atoms. Finally in 1961, the carbon atom was selected as the reference atom. In this scale, the relative mass a carbon atom was accepted as 12. The relative atomic mass of one hydrogen atom compared to the carbon atom becomes $12 \times 1/12$, that is 1. The mass of one proton and of one neutron on the scale of relative atomic masses is approximately one.



Research

The relative atomic masses of some elements in the chart below are given. You have to find the relative atomic masses of the others.

Element	Atomic mass	Element	Atomic mass	Element	Atomic mass
Hydrogen	1	Oxygen	-	Phosphorus	-
Helium	4	Fluorine	19	Sulphur	32
Lithium	7	Neon	20	Chlorine	35.5
Beryllium	9	Sodium	-	Argon	-
Boron	11	Magnesium	24	Potassium	-
Carbon	12	Aluminium	-	Calcium	40
Nitrogen	14	Silicon	28		

Today, we have highly accurate methods for measuring the mass of an atom directly. Hence, instead of relative mass, Unified Mass has now been accepted as the unit of atomic mass. It is called Dalton. Its symbol is 'u'. $1\text{u} = 1.66053904 \times 10^{-27}\text{ kg}$

Chemical symbols of elements



Can you tell?

1. How is an element represented in Chemistry?
2. Write down the symbols of the elements you know.
3. Write down the symbols for the following elements : antimony, iron, gold, silver, mercury, lead, sodium.

Dalton used certain signs to represent elements. For example $\odot$ for hydrogen, $\odot$ for copper. Today we use the symbols determined by IUPAC (International Union of Pure and Applied Chemistry). These are official names and symbols and are used all over the world. The current method of choosing chemical symbols is based on the method invented by Berzelius. According to this method the symbol of an element is either the first letter or the first and second/another specific letter in its name. Of the two letters, the first is written as capital letter and the second is small.

Molecules of elements and compounds

Atoms of some elements such as helium, neon have independent existence. It means that these elements are in a mono-atomic molecular state. Sometimes, two or more atoms of an element combine to form molecules of that element. Such elements are in a poly-atomic molecular state. For example, the elements oxygen, nitrogen are in a diatomic molecular state as O_2 , N_2 respectively. When atoms of different elements combine with each other, the molecules of compounds are formed. In other words, compounds are formed by chemical attraction between different elements.



Make a list and discuss

Make a list of elements that exist in mono-atomic and diatomic molecular states.

Molecular mass and the concept of mole

Molecular mass

The molecular mass of a substance is the sum of the atomic masses of all the atoms in a single molecule of that substance. Like atomic mass, molecular mass is also expressed in the unit Dalton (u).

How to deduce the molecular mass of H_2O ?

Molecule	Constituent elements	Atomic mass u	Number of atoms in the molecule	Atomic mass $\times$ number of atoms	Mass of the constituents u
H_2O	Hydrogen	1	2	1×2	2
	Oxygen	16	1	16×1	16
Molecular mass = sum of constituent atomic masses (Molecular mass of H_2O) = (Atomic mass of H) $\times$ 2 + (Atomic mass of O) $\times$ 1					Molecular Mass 18



Use your brain power !

Following are atomic masses of a few elements in Dalton and the molecular formulae of some compounds. Deduce the molecular masses of those compounds.

Atomic masses $\rightarrow$ H(1), O(16), N(14), C(12), K(39), S(32) Ca(40), Na(23), Cl(35.5), Mg(24), Al(27)

Molecular formulae $\rightarrow$ NaCl, MgCl_2 , KNO_3 , H_2O_2 , AlCl_3 , Ca(OH)_2 , MgO, H_2SO_4 , HNO_3 , NaOH

Mole



Try this

1. Weigh mass of single grains of *tur dal*, *masoor dal* and *chana dal* on a balance. What was your experience?
2. Weigh mass of 10g of *Tur dal*, *Masoor dal* and *Chana dal* each and count the number of *dal* grains of each kind in the weighed portions. Is the number the same for all or different?
3. Draw a line sketch on a paper. Paint it by placing/pasting the *dal* grains in the sequence *Tur*, *Masoor* and Bengal gram. You will need equal number of *dal*-grains of the three kinds. Count that number and also find out how many dozens of the respective *dal*-grains were required. Therefrom deduce the number of grams of *tur dal*, *masoor dal* and *chana dal* required for painting the picture.
4. What inference will you draw about the mass of an equal number of its different *dal* grains and the number of its *dal* grains in equal mass of different *dals*?



Think about it

How much of wheat, jowar and bajra seed is required for sowing one acre of land? Can these masses be correlated to the number of grains of the respective cereals they contain?



Can you tell?

1. Is it possible to weigh mass of one molecule using a weighing balance?
2. Will the number of molecules be the same in equal quantities of different substances?
3. If we want equal number of molecules of different substances, will it work to take equal masses of those substances?

When elements and compounds take part in chemical reactions, it is their atoms and molecules that react with each other, and therefore it is necessary to know the numbers of their atoms and molecules. However, while carrying out a chemical reaction it, is convenient to measure out quantities that can be handled instead of counting the numbers of atoms and molecules. The concept of 'mole' is useful for this purpose.

A mole is that quantity of a substance whose mass in grams is equal in magnitude to the molecular mass of that substance in Daltons. Thus, the molecular mass of oxygen is 32u, and therefore 32g oxygen is 1mole of oxygen. The molecular mass of water is 18u. Therefore, 18g of water make 1 mole of water.

1 mole of a compound is the mass of that substance in grams equal in magnitude to its molecular mass. The SI unit is mol.

$$\text{Number of moles of a substances (n)} = \frac{\text{Mass of substance in grams}}{\text{Molecular mass of substance}}$$

Avogadro's number

The number of molecules in one mole of any substance is constant. The Italian scientist Avogadro did fundamental work in this context. Therefore this number is called Avogadro's number and is denoted by the symbol N_A . Later scientists demonstrated experimentally that the value of Avogadro's number is 6.022×10^{23} . A mole of any substance stands for 6.022×10^{23} molecules. Just as a dozen has 12 items, a century has 100 or a gross has 144, a mole means 6.022×10^{23} . For example, a mole of water, that is, 18g of water contains 6.022×10^{23} molecules of water.

How many molecules are there in 66 g of CO_2 ?

Method : molecular mass of CO_2 is 44.

$$\text{Number of moles in the given } \text{CO}_2 (n) = \frac{\text{Mass of } \text{CO}_2 \text{ in grams}}{\text{Molecular mass of } \text{CO}_2} = \frac{66}{44}$$

$$\therefore n = 1.5 \text{ mol}$$

$$\therefore 1 \text{ mol of } \text{CO}_2 \text{ contains } 6.022 \times 10^{23} \text{ molecules.}$$

$$\therefore 1.5 \text{ mol } \text{CO}_2 \text{ contains } 1.5 \times 6.022 \times 10^{23} \text{ molecules} = 9.033 \times 10^{23} \text{ molecules}$$

**1 dozen bananas
are 12 bananas**



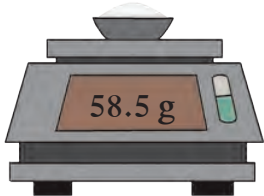
**144 notebooks make
1 gross of note books**



**1 mole common salt =
 6.022×10^{23} molecules**



1 mole water = 6.022×10^{23} molecules



4.3 One mole (Avogadro number)



Use your brain power !

1. How many molecules of water are there in 36g water?
2. How many molecules of H_2SO_4 are there in a 49g sample?



Always remember

1. The number of molecules in a given quantity of a substance is determined by its molecular mass.
2. The number of molecules in equal masses of different substances are different.
3. One mole quantities of different substances have different masses measured in grams.

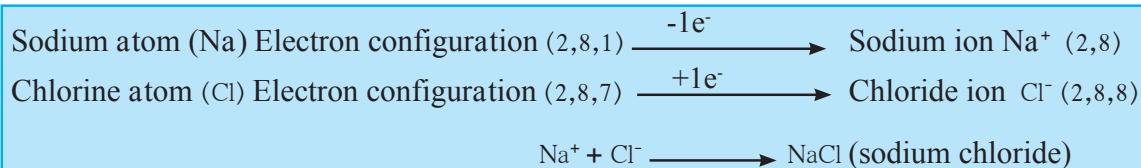
Valency



Can you recall?

1. Determine the valencies of H, Cl, O and Na from the molecular formulae H_2 , HCl, H_2O and NaCl.
2. What is the type of chemical bond in NaCl and $MgCl_2$?

The capacity of an element to combine is called its valency. The valency of an element is indicated by a specific number. It is the number of chemical bonds formed by one atom of that element with other atoms. In the 18th and 19th century, the laws of chemical combination were used to find out the valencies of elements. In 20th century, the relationship of the valency of an element with its electronic configuration was recognised.



A sodium atom gives away one electron and a cation of sodium is formed, hence, the valency of sodium is one. A chlorine atom takes up one electron and forms an anion of chlorine (chloride) and thus, the valency of chlorine is 1. After the give and take of electrons is over, the electronic configuration of both the resulting ions has a complete octet. Due to the attraction between the unit opposite charges on the two ions, one chemical bond is formed between Na^+ and Cl^- and the compound NaCl is formed.

Thus, a sodium atom has the capacity to give away one electron while a chlorine atom has the capacity to take up one electron. This means that the valency of both the elements sodium and chlorine is 1. From this the electronic definition of valency is as follows : “The number of electrons that an atom of an element gives away or takes up while forming an ionic bond, is called the valency of that element.”

Science capsule

Positively charged ions are called **cations** while negatively charged ions are called **anions**. For example, $MgCl_2$ contains Mg^{++} and Cl^- as cation and anion respectively.

The electrons present in the outermost orbit of an atom are called valence electrons.



Use your brain power !

How will the compounds, $MgCl_2$ and CaO be formed their elements?

The number of electrons that are given away or taken up is always a whole number. Therefore, valency is always a whole number.

The National Chemical Laboratory (NCL), a unit of the CSIR, was established in 1950. Its objectives are to conduct research in the various branches of chemistry, to aid industry and to develop new technology with a view to making profitable use of the country's natural resources. The Laboratory conducts research in fields such as biotechnology, nanotechnology, catalysis, drugs, instrumentation, agro chemicals, plant tissue culture and polymer science.

Complete the following chart.

Element	Atomic number	Electron configuration	Valence electrons	Valency
Hydrogen	1	1	1	1
Helium	2	2	2	0
Lithium		2,1		
Beryllium	4			2
Boron	5	2,3		
Carbon		2,4	4	
Nitrogen	7			3
Oxygen		2,6	6	
Fluorine	9		7	
Neon	10			
Sodium		2,8,1	1	1
Magnesium	12		2	
Aluminium	13	2,8,3		
Silicon	14		4	

Some elements that exhibit variable valency

Element	Symbol	Valency	Ion	Nomenclature
Copper	Cu	1 and 2	Cu ⁺ Cu ²⁺	Cuprous Cupric
Mercury	Hg	1 and 2	Hg ⁺ Hg ²⁺	Mercurous Mercuric
Iron	Fe	2 and 3	Fe ²⁺ Fe ³⁺	Ferrous Ferric

Variable valency

Under different conditions the atoms of some elements give away or take up different numbers of electrons. In such cases those elements exhibit more than one valency. This property of elements is called variable valency.



Always remember

Iron (Fe) exhibits the variable valencies 2 and 3. Therefore iron forms two compounds with chlorine FeCl₂ and FeCl₃.



Research

1. Find out some more elements which have variable valency.
2. Find out the compounds of the above elements which have variable valency.

Radicals



Complete the table.

Write down the cations and anions obtained from the compounds in the following chart.

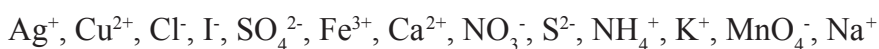
Base	Cation	Anion	Acid	Cation	Anion
NaOH			HCl		
KOH			HBr		
Ca(OH) ₂			HNO ₃		

Compounds with ionic bonds have two constituents. These are cations (positively charged ions) and anions (negatively charged ion). They take part independently in chemical reaction and are therefore, called radicals. It is seen from the above chart that different bases such as NaOH, KOH are formed when various cationic radicals are paired with the different anionic radical, hydroxide. Hence the cationic radicals are also called basic radicals. Different bases are distinguished from each other by the basic radicals in them. On the other hand, different acids, such as HCl, HBr are formed when different anionic radicals are paired with the cationic radical H^+ . Therefore, the anionic radicals are called acidic radicals. The difference in the composition of various acids becomes clear from the acidic radicals present in them.



Can you tell?

Which are the basic radicals and which are the acidic radicals among the following?

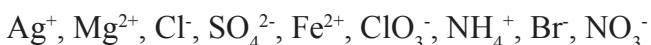


Generally, basic radicals are formed by removal of electrons from the atoms of metals, such as Na^+ , Cu^{2+} . But there are some exceptions, such as NH_4^+ . Similarly, the acidic radicals are formed by adding electrons to the atoms of non-metals, such as Cl^- , S^{2-} . But there are some exceptions like MnO_4^- .



Use your brain power !

Classify the following radicals into two types. While doing this use a criterion other than the sign of the electronic charge on them.



Monoatomic radicals such as Na^+ , Cu^+ , Cl^- are called simple radicals. When a radical is a group of atoms carrying charge, such as SO_4^{2-} , NH_4^+ , it is called composite radical. The magnitude of charge on any radical is its valency.

Chemical formulae of compounds : A recapitulation.

The characteristic of a compound formed by ionic bonds is that its molecule has two parts. These are a cation and an anion, that is, a basic radical and an acidic radical. These two parts are oppositely charged. The force of attraction between them constitutes the ionic bond. The name of an ionic compound contains two words. The first word is the name of the cation and the second is the name of the anion. For example, the formula of a compound such as sodium chloride is written by writing the symbol of the cation on the left and adjoined to it on the right is the symbol of the anion. The charges are not shown though the number of the ions is written as a subscript on the right of the symbol of the ion. In the case of composite radical the number is written as subscript outside the bracket enclosing its symbol. This number is obtained easily by the method of cross multiplication of the valencies. The steps for writing a chemical formula are shown on the next page.

Using ICT

To study 'Measurement of Matter' further and for additional information, visit the websites given alongside.

Prepare a spreadsheet showing the atomic mass of elements, molecular masses, electronic configurations and valencies.

Website for more information

www.organic.chemistry.org

www.masterorganicchemistry.com

www.rsc.org.learnchemistry

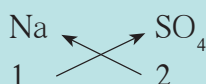
Step 1 : To write the symbols of the radicals. (Basic radical on the left.)



Step 2 : To write the valency below the respective radical.



Step 3 : To cross-multiply as shown by the arrows the number of the radicals.



Step 4 : To write down the chemical formula of the compound.



In order to write the chemical formulae of compounds, it is necessary to know the valency of the various radicals. The names and symbols along with the charge of common radicals are given in the chart below.

Ions/Radicals

Basic Radicals

H ⁺	Hydrogen	Al ³⁺	Aluminium
Na ⁺	Sodium	Cr ³⁺	Chromium
K ⁺	Potassium	Fe ³⁺	Ferric
Ag ⁺	Silver	Au ³⁺	Gold
Hg ⁺	Mercurous	Sn ⁴⁺	Stannic
Cu ⁺	Cuprous	NH ₄ ⁺	Ammonium
Cu ²⁺	Cupric/Copper		
Mg ²⁺	Magnesium		
Ca ²⁺	Calcium		
Ni ²⁺	Nickel		
Co ²⁺	Cobalt		
Hg ²⁺	Mercuric		
Mn ²⁺	Manganese		
Fe ²⁺	Ferrous (Iron II)		
Sn ²⁺	Stannous		
Pt ²⁺	Platinum		

Acidic Radicals

H ⁻	Hydride	MnO ₄ ⁻	Permanganate
F ⁻	Fluoride	ClO ₃ ⁻	Chlorate
Cl ⁻	Chloride	BrO ₃ ⁻	Bromate
Br ⁻	Bromide	IO ₃ ⁻	Iodate
I ⁻	Iodide	CO ₃ ²⁻	Carbonate
O ²⁻	Oxide	SO ₄ ²⁻	Sulphate
S ²⁻	Sulphide	SO ₃ ²⁻	Sulphite
N ³⁻	Nitride	CrO ₄ ²⁻	Chromate
		Cr ₂ O ₇ ²⁻	Dichromate
OH ⁻	Hydroxide	PO ₄ ³⁻	Phosphate
NO ₃ ⁻	Nitrate		
NO ₂ ⁻	Nitrite		
HCO ₃ ⁻	Bicarbonate		
HSO ₄ ⁻	Bisulphate		
HSO ₃ ⁻	Bisulphite		



Books are my friends

Essentials of Chemistry, The Encyclopedia of Chemistry, Science and Technology Dictionary.

**Use your brain power !**

Using the chart of ions/radicals and the cross-multiplication method, write the chemical formulae of the following compounds : Calcium carbonate, Sodium bicarbonate, Silver chloride, Calcium hydroxide, Magnesium oxide, Ammonium phosphate, Cuprous bromide, Copper sulphate, Potassium nitrate, Sodium dichromate.

Exercises**1. Give examples.**

- Positive radicals
- Basic radicals
- Composite radicals
- Metals with variable valency
- Bivalent acidic radicals
- Trivalent basic radicals

2. Write symbols of the following elements and the radicals obtained from them, and indicate the charge on the radicals.

Mercury, potassium, nitrogen, copper, sulphur, carbon, chlorine, oxygen

3. Write the steps in deducing the chemical formulae of the following compounds.

Sodium sulphate, potassium nitrate, ferric phosphate, calcium oxide, aluminium hydroxide

4. Write answers to the following questions and explain your answers.

- Explain the monovalency of the element sodium.
- M is a bivalent metal. Write down the steps to find the chemical formulae of its compounds formed with the radicals : sulphate and phosphate
- Explain the need for a reference atom for atomic mass. Give some information about two reference atoms.
- What is meant by Unified Atomic Mass?
- Explain with examples what is meant by a 'mole' of a substance.

5. Write the names of the following compounds and deduce their molecular masses.

Na_2SO_4 , K_2CO_3 , CO_2 , MgCl_2 , NaOH , AlPO_4 , NaHCO_3

6. Two samples 'm' and 'n' of slaked lime were obtained from two different reactions. The details about their composition are as follows:

'sample m' mass : 7g

Mass of constituent oxygen : 2g

Mass of constituent calcium : 5g

'sample n' mass : 1.4g

Mass of constituent oxygen : 0.4g

Mass of constituent calcium : 1.0g

Which law of chemical combination does this prove? Explain.

7. Deduce the number of molecules of the following compounds in the given quantities.

32g oxygen, 90g water, 8.8g carbon dioxide, 7.1g chlorine.

8. If 0.2 mol of the following substances are required how many grams of those substances should be taken?

Sodium chloride, magnesium oxide, calcium carbonate

Project :

Prepare models of various radicals using cardboard, small circular magnets and araldite. From these make models of molecules of various compounds.

